

Bias Mitigation in Federated Learning for Edge Computing

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Federated learning (FL) is a distributed machine learning paradigm that enables data owners to collaborate on training models while preserving data privacy. As FL effectively leverages decentralized and sensitive data sources, it is increasingly used in ubiquitous computing including remote healthcare, activity recognition, and mobile applications. However, FL raises ethical and social concerns as it may introduce bias with regard to sensitive attributes such as race, gender, and location. Mitigating FL bias is thus a major research challenge. In this paper, we propose ASTRAL, a novel bias mitigation system for FL. ASTRAL provides a novel model aggregation approach to select the most effective aggregation weights to combine FL clients' models. It guarantees a predefined fairness objective by constraining bias below a given threshold while keeping model accuracy as high as possible. ASTRAL handles the bias of single and multiple sensitive attributes and supports all bias metrics. Our comprehensive evaluation on seven real-world datasets with three popular bias metrics shows that ASTRAL outperforms state-of-the-art FL bias mitigation techniques in terms of bias mitigation and model accuracy. Moreover, we show that ASTRAL is robust against data heterogeneity and scalable in terms of data size and number of FL clients. ASTRAL's code base is publicly available.

CCS Concepts: • **Computing methodologies** → **Distributed computing methodologies**; **Distributed artificial intelligence**.

Additional Key Words and Phrases: Federated Learning, Bias, Fairness

ACM Reference Format:

Yasmine Djebrouni, Nawel Benarba, Ousmane Touat, Pasquale De Rosa, Sara Bouchenak, Angela Bonifati, Pascal Felber, Vania Marangozova, and Valerio Schiavoni. 2023. Bias Mitigation in Federated Learning for Edge Computing. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 4, Article 157 (December 2023), 35 pages. <https://doi.org/10.1145/3631455>

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2474-9567/2023/12-ART157 \$15.00

<https://doi.org/10.1145/3631455>

1 INTRODUCTION

Ubiquitous computing systems, also known as pervasive computing systems, are today seamlessly integrated into our everyday environment and activities, providing intelligent services and applications such as disease diagnosis [57], activity recognition [64], traffic management [14], self-driving car monitoring [5], etc. Federated Learning (FL) [45], an emerging distributed machine learning paradigm, is gaining attention as a promising approach to intelligent services in ubiquitous computing systems. It allows multiple data owners referred to as clients to collaboratively build a machine learning model by learning from each other's data while keeping data private at their premises. This approach offers a number of advantages for ubiquitous systems over traditional centralized machine learning, including increased privacy, reduced data transfer, and the ability to learn from diverse and geographically distributed data sources. In particular, FL has great promise for improving the capabilities of ubiquitous computing systems in critical domains where data is sensitive, such as disease diagnosis [7], smart governance [54], and smart security [32].

Despite the advantages of FL, it is important to acknowledge and address the potential risks and challenges associated with its use. One key concern in FL models is the potential for model bias and unfairness. Bias or unfairness are present in a system whose decisions are prejudicial against users based on their sensitive attributes. Sensitive attributes capture sensitive and demographic information that is protected by law and may include, among others, religion, race, gender, and age [19, 71]. Instances of unfair machine learning algorithms in the real world are abundant. For example in [60, 74], the authors show that health sensors like oximeters consistently misclassify individuals of color as they have mostly been tested on white populations. Similarly, voice recognition models may exhibit higher accuracy for men than for women due to the differences in vocal characteristics between genders. Women's voices are generally higher in pitch and thinner, making it more challenging for accelerometers to capture voice features compared to men's voices [63].

Bias in machine learning systems is mainly due to unbalanced, incomplete, flawed, or detrimental models and datasets [53]. For example, it can arise when the data used to train machine learning models is imbalanced and not representative, or when data is inherently unfair holding a biased ground truth against some population groups. In FL, bias can be caused by clients that inadvertently train biased models because of their biased local data and then propagate bias to the global aggregated model. In ubiquitous computing applications of FL, the emergence of novel sources of bias in conjunction with inherent data bias is a significant concern. Participating clients may indeed exhibit variability not only in their local data distributions but also in connectivity and resource availability. However, as this variability may reflect the demographic and socio-economic profiles of clients, client exclusion may reduce model representativeness and increase the likelihood of bias [74]. The potential harm of biased AI and ML techniques ignited several legislative actions, *e.g.*, the *National AI Initiative Act* in the US [66] and the *EU AI Act* [65], aiming at promoting fairness, accountability, and transparency in AI systems.

Bias represents a widely recognized concern that has been extensively explored in machine learning [8, 46]. In FL, there are different types of bias problems depending on the considered FL fairness notion. These notions include (i) collaborative fairness (*e.g.* CFFL [42], and FLI [75]) which aims to reward highly-contributing clients with better models in terms of performance compared to low-contributing clients, (ii) performance fairness (*e.g.* Agnostic Federated Learning [49], FedMGDA+ [30], FCF [81], and FedCHAR [38]) which aims to ensure a performant FL model for all FL clients, and (iii) group fairness, where data records belong to different groups, and where the objective is to prevent discrimination between groups in the performance or outcome of the FL model. In this paper, we focus on mitigating bias considering the third type of fairness notion.

Bias mitigation techniques used to achieve group fairness in classical (*i.e.* centralized) ML are incompatible with the privacy constraints of federated learning. If several recent research proposals consider the problem of bias in FL [11, 16, 20, 37, 76, 80], none but FCFL [11] guarantees a predetermined bias objective for the resulting model. These solutions are therefore incompatible with legal requirements regarding bias and fairness in various

application domains. Furthermore, most works are restricted to a single sensitive attribute. However, handling bias for a single sensitive attribute may deepen bias regarding other sensitive attributes.

To address the bias mitigation problem in FL, in this paper we propose ASTRAL, a bias mitigation framework that constrains bias below a given threshold, while maintaining FL model accuracy as high as possible. ASTRAL proposes a novel and self-corrective FL aggregation method, which reweighs FL clients' local models contributions based on their impact on the global FL model's bias and accuracy. In contrast to its competitors, ASTRAL allows dealing with multiple sensitive attributes at a time, and supports bias mitigation with regard to any given bias metric. To showcase ASTRAL's results in terms of model bias, accuracy, scalability, and robustness to clients heterogeneity, we conduct an extensive evaluation across seven widely used datasets. Our findings confirm that ASTRAL outperforms existing FL bias mitigation approaches.

In this paper, we make the following main contributions:

- We design and implement ASTRAL, a bias mitigation framework that handles bias with respect to multiple sensitive attributes and supports multiple bias metrics, thus adapting bias mitigation to diverse practical use cases.
- We show that ASTRAL effectively works on multiple attributes as opposed to its competitors.
- We enable ASTRAL to effectively mitigate bias by ensuring it remains below a given threshold and maintaining model accuracy as high as possible.
- We evaluate ASTRAL with seven widely used datasets and various FL distributed and heterogeneous settings, and compare it against three other FL bias mitigation techniques.
- We show that ASTRAL is scalable and achieves bias and accuracy objectives independently from the dataset size, the number of participating clients, and the client selection ratios.
- We show that ASTRAL is efficient and achieves its objectives independently from the degree of heterogeneity of participating clients.
- We assess that ASTRAL is a practical FL bias mitigation solution, and show that it is more communication-efficient than its competitors, as it integrates optimization objectives by design and, thus, fulfills these goals at early training stages.
- We make the ASTRAL software prototype and used datasets available: <https://github.com/FL-Bias/ASTRAL>

Paper Roadmap. The rest of the paper is organized as follows. In §2, we introduce essential background information on FL and bias and survey related work on bias mitigation in FL. In §3 we state the problem that motivates our own work. The problem formulation and the design principles of our solution ASTRAL are presented in §4. In §5, we carry extensive empirical evaluations to validate our proposal and compare it to other FL bias mitigation techniques. Finally, we draw our conclusions in §6.

2 BACKGROUND AND RELATED WORK

In this section, we start by providing the necessary background on FL and on bias in FL. We illustrate these notions with an example of a FL ubiquitous system. Finally, we cover related work.

2.1 Federated Learning

Federated Learning (FL) allows data owners (or clients) to collaboratively train an ML model without exchanging their data. The training process is managed by a central server and is organized in rounds. At each round, the clients and the server work on copies of the same ML model defined by a set of parameters $\theta \in \mathbb{R}^d$, where d is the number of the model's parameters. The N clients, *e.g.*, $c_1, c_2, \dots, c_N$, train the ML model on their local data and produce new models, *i.e.*, local models. Each client c_k sends the parameters $\theta_k \in \mathbb{R}^d$ of its local model to the server. The server aggregates the clients' inputs and produces a new version of the model, *i.e.*, the global model, to be used in the next round.

In FL systems, data is usually distributed among clients in a non-IID manner (non-identically and independently distributed) [28]. The local data of client c_k is denoted U_k and contains n_k samples of data. U_k is referred to as the *local distribution* of client k . The union of all clients' data is denoted $\mathcal{U}$ and is referred to as the *global distribution*. In case of classification learning, we denote a training sample of data from client c_k by $\{\bar{x}_i^k, y_i^k\}, i \in \{1, \dots, n_k\}$, where $\bar{x}_i^k$ is the vector of its attributes' values, and y_i^k is its class label. y_i^k is referred to as the ground truth. We use $\mathcal{X}$ and $\mathcal{Y}$ to denote respectively the spaces for the values respectively of attributes and of class labels.

The classical FL aggregation method for combining clients' models is FedAvg [45]. It aggregates the clients' parameters to obtain the parameters of the global model. It uses Eq. 1 according to which the weight of a client's parameters is proportional to the quantity of its data. The output of the FL system is a classification model that predicts the class label value for an unlabeled data instance $\bar{x} \in \mathcal{X}$. The model's prediction for $\bar{x}$ is referred to as $\hat{y}$. The quality of the model's predictions is usually measured with metrics such as model accuracy, precision or recall.

$$\theta = \sum_{k=1}^{k=N} \frac{n_k}{|\mathcal{U}|} \cdot \theta_k \quad (1)$$

2.2 Bias in Federated Learning

Bias in FL is defined as the presence of prejudice or favoritism in a FL model outcome that disproportionately affects certain individuals based on their sensitive attributes such as race or gender [47]. Such bias leads to unfair outcomes and exacerbates existing social inequalities. It can result in a range of negative consequences, such as denying opportunities, resources, or services to certain individuals or groups. ML model bias with respect to sensitive attributes has been addressed in several studies in the emerging field of ethical machine learning [18, 22, 35]. These studies consider different bias concepts including statistical parity difference [18], equal opportunity difference [25] and discrimination index [80]. While these concepts differ somewhat, they all quantify bias by examining if and how the outcomes of the models depend on sensitive attributes. In the literature, these concepts are grouped under the term of *group fairness* [18]. Sensitive attributes are defined by law and include, among many others, race, age, gender, health situation, *etc.* [19, 71]. A sensitive attribute, denoted S_j , typically has two possible values that we denote by v_1^j and v_2^j . It separates data into two demographic groups. Usually, we refer to these groups as privileged and unprivileged groups. In fairness literature, the *privileged group* is a group that has historically been at a systemic advantage [43], while the *unprivileged group* has been disadvantaged. For example, in the United States, race has been a significant factor in determining access to resources and opportunities. White individuals have had more advantages than individuals from other racial groups, and are thus identified as the privileged group. However, it is important to note that the definition of privileged and unprivileged groups may vary depending on the context and the specific task at hand.

Let us now consider the ubiquitous use case of smart healthcare in which several hospitals use FL for activity recognition [52], as illustrated in Fig. 1. A central server aggregates into a global model the hospitals' local models that are trained using sensor data collected from their local patients. An imbalance in the collected data can result in a biased model against underrepresented (*e.g.* women) groups.

2.3 Bias Metrics

To quantify bias, a variety of metrics have been proposed [18, 25]. These metrics operate under the assumption that the sensitive attribute is binary (we explore in §5.1 the way we binarize sensitive attributes in our experiments). We denote the number of sensitive attributes as sa , and the model outcome variable, which we consider binary for simplicity, as $\hat{Y} \in \{0, 1\}$. The value 1 is referred to as the positive outcome, while 0 is referred to as the

negative outcome. In the following, we briefly detail the different bias metrics considered in our work. In practice, the choice of which metric to use depends on the specific context, the type of discrimination that needs to be addressed, and the desired outcome of the system.

Statistical Parity Difference (SPD) is a classical bias metric for binary classifiers used to measure the demographic parity fairness notion [18]. It measures the difference between the positive outcome rates for the two groups defined by the considered sensitive attribute. A model is fair and ensures demographic parity if its SPD is equal to zero. If the SPD is different than zero, it indicates that the model is more likely to assign positive predictions to one particular group. Given a dataset $\mathcal{U}$ and a sensitive attribute S_j , SPD_{S_j} is defined in Eq. 2, with $Pr_{\mathcal{U}}$ being the probability distribution of $\mathcal{U}$. SPD is used in systems interested only in the prediction outcome. For example, consider the use case of a mobile recruiting application [4]. Measuring the SPD with respect to attributes such as race or gender on the output of the hiring system can indicate whether the model is hiring candidates of certain demographic groups more than the others.

$$SPD_{S_j} = Pr_{\mathcal{U}}(\hat{Y} = 1 | S_j = v_1^j) - Pr_{\mathcal{U}}(\hat{Y} = 1 | S_j = v_2^j) \quad (2)$$

Equal opportunity Difference (EOD) is used to reflect the fairness notion of equality of opportunity [25]. Unlike SPD, EOD imposes constraints on the *correct* or *true* positive predictions [25]. Formally, EOD measures the difference between the correct positive predictions for the two groups defined by S_j , as presented in Eq. 3. It is more prone to be used in systems where not only the outcome but also the quality of prediction matter, such as identifying dangerous individuals from cameras' footage [3].

$$EOD_{S_j} = Pr_{\mathcal{U}}(\hat{Y} = 1 | S_j = v_1^j, Y = 1) - Pr_{\mathcal{U}}(\hat{Y} = 1 | S_j = v_2^j, Y = 1) \quad (3)$$

Discrimination Index (DI) quantifies the variation of the F1 score between the two groups defined by a sensitive attribute S_j [80]. The F1 score is a statistical measure that evaluates the quality of a binary classification model, and is equal to the harmonic mean of the recall and the precision measures. Formally, DI for a model θ with regard to the sensitive attribute S_j is calculated as in Eq. 4, where $F1_{S_j=v_i^j}(\theta)$ denotes the F1 score of the model θ computed using only data records from the group with $S_j = v_i^j$. DI is more prone to be used in systems where the prediction quality is of utmost concern such as healthcare applications [2]

$$DI_{S_j} = F1_{S_j=v_2^j}(\theta) - F1_{S_j=v_1^j}(\theta) \quad (4)$$

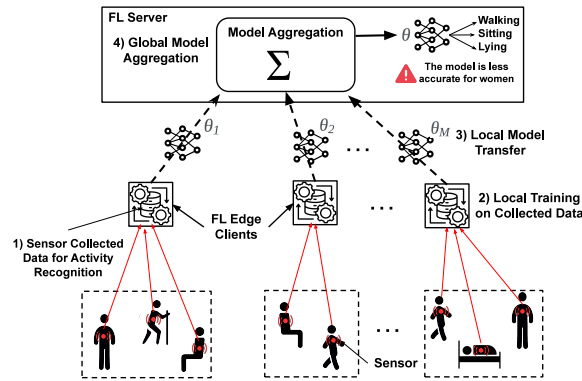


Fig. 1. Example of bias in ubiquitous FL systems – Healthcare and activity recognition

2.4 Related Work on FL Bias Mitigation

Bias is a well-known problem studied in ML [8, 46]. In FL, there are different categories of bias problems depending on the considered FL fairness notion. The first category of problems deals with (i) collaborative fairness which aims to reward clients with high contribution to the FL process with better models in terms of performance compared to low-contributing clients. The second category deals with (ii) performance fairness which aims to achieve a performant FL model for the different clients. The third category deals with (iii) group fairness, where the aim is to prevent any discrimination in the FL model against particular data groups based on their sensitive attributes.

Several works study the first type of bias problems [42, 75]. In [42], the authors introduce a reputation mechanism that ensures a correlation between the contributions of FL clients and their respective final model accuracies. In [75], the authors propose the FLI payoff-sharing scheme, which incentivizes FL data owners to contribute high-quality data to the FL system. Data owners who have contributed a large set of high-quality get rewarded by a higher share of revenues generated by the FL system.

Other works focus on the second category of bias problems, specifically performance fairness. For instance, AFL [49] ensures that the global model is optimized for any target distribution formed by a mixture of the client distributions by minimizing the worst convex loss combination. In a different approach, authors in [81] introduce a fair and communication-efficient FL selection protocol to ensure that the final model is representative of all the clients distributions. Additionally, [38] proposes FedCHAR, a personalized FL system that improves the accuracy and the fairness of model performance by exploiting the intrinsically similar relationship between FL clients. In this paper, we direct our attention to the third type of bias issue, namely group fairness.

Several FL bias mitigation techniques for group fairness exist, including client-side [1], and server-side methods [15, 35, 76–78]. Client-side techniques adjust local loss with bias penalties and reweigh client data based on sensitive attributes. However, this does not guarantee global bias mitigation since data distribution is heterogeneous among FL clients [20]. On the other hand, several works follow a server-side approach for FL bias mitigation [9, 11, 16, 20, 79, 80]. Among them, AgnosticFair [16] considers an approximate metric for the SPD, and applies a minimax gaming optimization to minimize the approximate under a given threshold. It assumes the presence of a single sensitive attribute in data [16] and requires exchanges of local bias information at each FL learning round. Another proposal is FCFL [11] which considers the SPD and EOD metrics. It solves a multi-objective constrained optimization problem to enforce that the global model respects the bias constraint on each client's distribution and ensures consistent accuracy across clients. Here, client's bias and loss information is exchanged at each FL round. Similarly to AgnosticFair, FCFL handles only single sensitive attribute bias.

FairFL [80] adopts a different approach with specific client selection based on clients' collected measurements. It selects at each FL round the best set of clients to optimize accuracy and bias approximate functions. The approximates are computed using clients' accuracy and bias measurements. FairFL considers the existence of multiple sensitive attributes but fails at reducing effectively the bias for each of the existing groups [80]. FairFed [20] adapts the clients' weights for the global aggregation to minimize SPD and EOD as much as possible, by applying on each client's weight a factor dependent on the client's bias. FairFed requires additional global statistical information, as well as local bias measurements from FL clients, and does not handle the existence of multiple sensitive attributes.

Finally, there exist works that explicitly address group fairness by addressing performance fairness for groups of FL clients clustered according to their sensitive attributes. Specifically, q -FFL [37], encourages a more equitable distribution of accuracy across FL clients. It reweighs the aggregate loss and assigns higher weights to devices with higher losses, resulting in reducing discrimination between FL clients and thus between demographic groups when the clients are clustered into groups with specific sensitive attributes. GIFAIR-FL [76] on the other hand offers an optimization-based approach to address group fairness with regard to model accuracy among groups

Table 1. Summary of state-of-the-art bias mitigation methods

System	Handling multiple sensitive attributes	Ensuring a bias threshold	Generic approach
Abay and al. [1]	✗	✗	✓
AgnosticFair [16]	✗	✗	✗
FCFL [11]	✗	✓	✗
FairFL [80]	✓	✗	✓
FairFed [20]	✗	✗	✓
GIFAIR-FL [76]	✓	✗	✗
q -FFL [37]	✓	✗	✗
ASTRAL	✓	✓	✓

of clients. This is achieved through the application of regularization methods that penalize variations in model accuracy between clients' groups. A particularity of q -FFL and GIFAIR-FL is that they reduce discrimination between demographic groups under the assumption that each FL client can be clearly assigned to a demographic group. However, if this assumption is not met, the used approach does not guarantee group fairness. In our paper, we refrain from making any assumptions regarding a client's affiliation with a specific demographic group. Specifically, clients can hold data records belonging to one demographic group or to different ones.

In summary, most existing bias mitigation methods target only single sensitive attribute bias. They are not generic as they do not consider all existent bias metrics. Moreover, related works require FL clients to share additional information. All this limits their application across different domains which have different needs in terms of bias guarantees. Table 1 compares the different bias mitigation methods including our solution. As far as we know, ASTRAL emerges as the only FL bias mitigation approach capable of handling multiple sensitive attributes. It ensures that the model bias remains below adjustable threshold while maximizing accuracy. It accommodates to any bias metric, and does not mandate extra client data sharing.

3 PROBLEM MOTIVATION

3.1 Why Bias Matters in Federated Learning

In federated learning, the presence of biased data at an FL client may lead to the propagation of bias to all the FL clients, including the "fair" ones, thus rendering the whole system biased. Indeed, clients that have unbiased local data, that allow them to train fair models, can "inherit" a biased global FL model, due to the biased FL clients, as shown in Fig. 2(a). This rises a problem, since classical FL protocols do not allow to select clients based on their data due to FL privacy constraints. To illustrate these phenomena, we implement an activity recognition FL scenario that uses motion data collected from wearable sensors from several patients¹. Here, we consider gender as sensitive attribute and DI as the bias metric. We compare the DI of the following two models. The first model is trained using classical centralized machine learning on a FL client denoted c_1 , with a balanced data subset of men and women. The second model is trained collaboratively by c_1 and other FL clients holding the remaining data using the classical FL protocol FedAvg. We present the corresponding bias values in Fig. 2(b). We can observe that FL induces a much higher bias on the resulting model (right bar). Indeed, the unbiased FL client c_1 inherited bias from the other FL clients.

3.2 On the Importance of Maintaining Bias Below a Threshold

Minimizing the gap between groups without establishing a specific threshold may lead to two potential outcomes: either a model with reduced bias but still exhibiting more bias than desired, or a model with no bias but

¹These experiments use the ARS dataset (see §5.1), executed on our experimental environment (see §5.2).

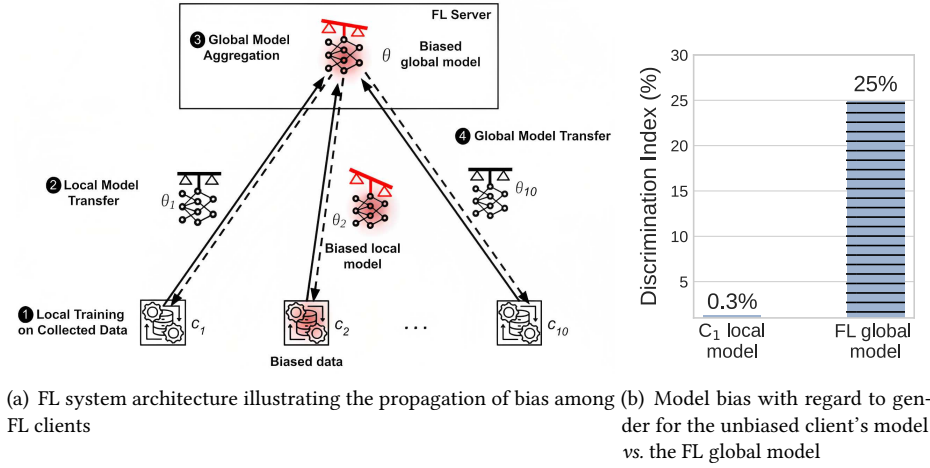


Fig. 2. Bias propagation in FL

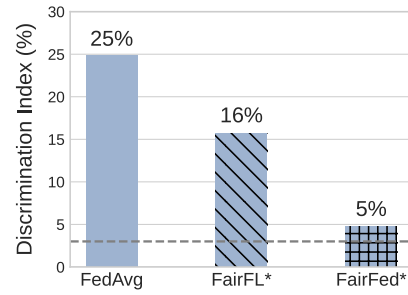
compromised accuracy. The latter occurs because, in the pursuit of mitigating bias to the greatest extent, the model may excessively compensate for certain groups without considering the underlying ground truth. Moreover, fixing a threshold in a bias mitigation system ensures that the system is consistent with legal requirements. Indeed, in many real-world scenarios, quotas are defined to ensure that certain groups are not unfairly disadvantaged. For example in the United States, there exist a rule known as the 80% rule, or the four-fifths rule, that is imposed on companies in their hiring process to fight against hiring discrimination [31]. It ensures that the quotient of the hiring rate of the unprivileged group on the privileged group's is no less than 80%.

3.3 Existing FL Bias Mitigation Methods Are Not Able to Limit Bias Below a Threshold

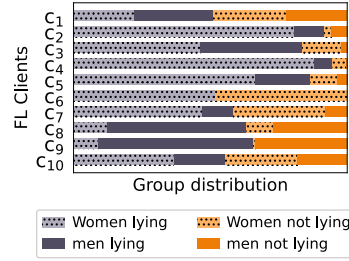
To investigate the effectiveness of FL bias mitigation methods, we consider the same activity recognition FL scenario as above, initially without any bias mitigation, and subsequently, using two state-of-the-art bias mitigation techniques. We evaluate the bias (DI) of the learned FL model with regard to gender and report the results in Fig. 3(a). It reveals that the FL model created without any bias mitigation methods has a DI of 25%, surpassing the threshold 3%. We then apply two bias mitigation techniques: FairFL* and FairFed* (described in details in §5.3). FCFL (see §5.3), another bias mitigation technique cannot be applied since it does not support bias mitigation with regard to DI. We can see from Fig. 3(a) that none of these methods succeeds in reducing bias below the typical 3% bias threshold within the setting shown in Fig. 3(b). This illustrates the fact that bias mitigation represents a real challenge in FL.

3.4 Why Not Handling Multiple Sensitive Attributes May Induce More Bias

FL applications typically use data that contains multiple attributes that can be considered sensitive. For example, data collected from mobile applications may include information on an individual's gender, race, and age, while data collected from sensors and other ubiquitous devices may include attributes revealing the region and socio-economic information, among others. Therefore, data in general is characterized by the presence of several sensitive attributes. Although several works have been conducted to mitigate bias in FL, most of them focus on mitigating bias with regard to only one sensitive attribute at a time. This may be problematic, as optimizing a model to be unbiased with regard to one sensitive attribute does not guarantee bias mitigation for other sensitive



(a) FL model bias with regard to gender for the activity recognition FL task: without bias mitigation vs. state-of-the-art bias mitigation



(b) Non-IID client data distribution of the activity recognition FL task

Fig. 3. Existing FL bias mitigation methods are not able to achieve the defined threshold

Table 2. Bias mitigation with regard to single sensitive attribute age – Adult dataset

Systems	$SPD_{age}(\%)$	$SPD_{race}(\%)$	$SPD_{gender}(\%)$
FedAvg	13	4	13
FCFL	1	9	19

Table 3. Bias mitigation with regard to single sensitive attribute race – MEPS dataset

Systems	$SPD_{race}(\%)$	$SPD_{gender}(\%)$
FedAvg	5	8
FCFL	2	10

attributes. Even worse, mitigating bias with regard to one sensitive attribute may exacerbate bias with regard to the other sensitive attributes. We have observed this phenomenon when applying the state-of-the-art bias mitigation method FCFL on several datasets considering the SPD metric. FCFL handles single attribute bias mitigation only. Table 2 and Table 3 show bias measurements collected from the FL model resultant from this method and obtained using respectively the dataset Adult that contains three sensitive attributes, and MEPS that contains two sensitive attributes. We report the model bias for the two datasets with regard to one sensitive attribute that was considered during the mitigation (bias column at left), and also bias with regard to the other sensitive attributes that exist in the dataset (bias columns at right). As we can see, although FCFL mitigates model bias with regard to age for the Adult dataset, and with regard to race for the MEPS, it exacerbates the model bias with regard to the remaining sensitive attributes. Thus it is important to design bias mitigation techniques that handle bias mitigation for all the sensitive attributes that exist in data.

4 THE ASTRAL SYSTEM

In this section we present ASTRAL, a federated learning framework for bias mitigation. First, we define formally the problem of bias mitigation in FL in §4.1. Then, we present an overview of ASTRAL to solve the formulated problem in §4.2. We describe in detail the design principles of our solution in §4.3. Finally, in §4.4, we provide analytical insights of ASTRAL.

4.1 Problem Formulation

The problem of bias mitigation in FL consists in designing an FL system that produces FL models meeting a predetermined bias threshold, and this for all the sensitive attributes existing in data simultaneously. As respecting bias constraints should not come at the expense of accuracy, the accuracy of the resulting FL model

should be kept as high as possible. The target distribution in FL systems is the global distribution $\mathcal{U}$, thus the bias and the accuracy objective must be ensured for the global distribution $\mathcal{U}$ [45].

More formally, the defined problem of FL bias mitigation can be formulated as a constrained optimization problem as follows. Given an FL model with parameters θ , the accuracy $A(\theta)$ of the model should be maximized (Eq. 5a), while the model bias $\beta_{S_j}(\theta)$ with regard to each sensitive attribute S_j does not exceed a predefined threshold ϵ (Eq. 5b). Here β is used to denote any bias metric (SPD, EOD, DI, etc. (See §2.2)).

$$\max_{\theta \in \mathbb{R}^d} A(\theta) \quad (5a)$$

$$\text{s.t } |\beta_{S_j}(\theta)| \leq \epsilon, \forall S_j \in \{S_1 \dots S_{sa}\} \quad (5b)$$

The global model parameters θ can be expressed as a linear combination of the local clients' parameters $\theta_1, \theta_2 \dots \theta_N$, and this according to the aggregation formula in Eq. 1. If $\theta_k \in \mathbb{R}^d$ are the local parameters for client c_k , and $w_k \in \mathbb{R}$ the weight assigned to this client for the aggregation, the global model parameters produced in a FL round are computed according to Eq. 6.

$$\theta = \sum_{k=1}^{k=N} w_k \cdot \theta_k \quad (6)$$

To simplify Eq. 6, we represent clients' aggregation weights as the vector $\bar{w} = (w_1 \dots w_N)$, and clients' parameters as the $N \times d$ matrix Θ (Eq. 7). In Θ , row k contains the d parameters for client c_k , d being the number of parameters of the learning model.

$$\Theta = \begin{pmatrix} \theta_1^1 & \dots & \theta_1^d \\ \dots & \dots & \dots \\ \theta_N^1 & \dots & \theta_N^d \end{pmatrix} \quad (7)$$

Thus, we have $\theta = \bar{w} \cdot \Theta$. Therefore, the problem formulated by Eq. 5a and Eq. 5b can be expressed using the clients' weights $\bar{w}$ and thus formulated as follows:

$$\max_{\bar{w} \in \mathbb{R}^N} A(\bar{w} \cdot \Theta) \quad (8a)$$

$$\text{s.t } |\beta_{S_j}(\bar{w} \cdot \Theta)| \leq \epsilon, \forall S_j \in \{S_1 \dots S_{sa}\} \quad (8b)$$

In order to solve the constrained maximization problem formulated by Eq. 8a and Eq. 8b, our solution finds the best real number aggregation weights $w_1, w_2 \dots w_N$.

Bias metrics in Eq. 8b are known to be non-convex functions [72]. As a result, the considered problem is also non-convex. Thus, the considered problem is computationally intractable using exact methods, *i.e.* it can not be solved efficiently in polynomial time [23]. Furthermore, the exploration of the search space becomes intractable when considering trivial optimization methods like grid search. Indeed, as our search space is potentially infinite ($\bar{w} \in \mathbb{R}^N$), we have to resort to more sophisticated methods to more efficiently explore the search space. Thus, to find the best weights vector $\bar{w}$, we apply a black-box optimization method, which does not require a specific type (convexity, continuity, *etc.*) of the studied function [56]. The black-box optimization relies on the exploration of different candidates of aggregation weights through a metaheuristic-based algorithm. The algorithm guides efficient and non-exhaustive exploration of the search space and outputs the best weights that optimize the objectives at the end of the learning.

4.2 Overview of ASTRAL

Fig. 4 presents an overview of the architecture of ASTRAL, which intervenes at each FL round at the FL server-side after receiving the client's parameters and before the aggregation step. Specifically, ASTRAL applies the pipeline described in Fig. 5 to calculate the weights used for aggregating the FL client model updates in a way that ensures the aggregated global model bias remains below a fixed threshold while maximizing accuracy.

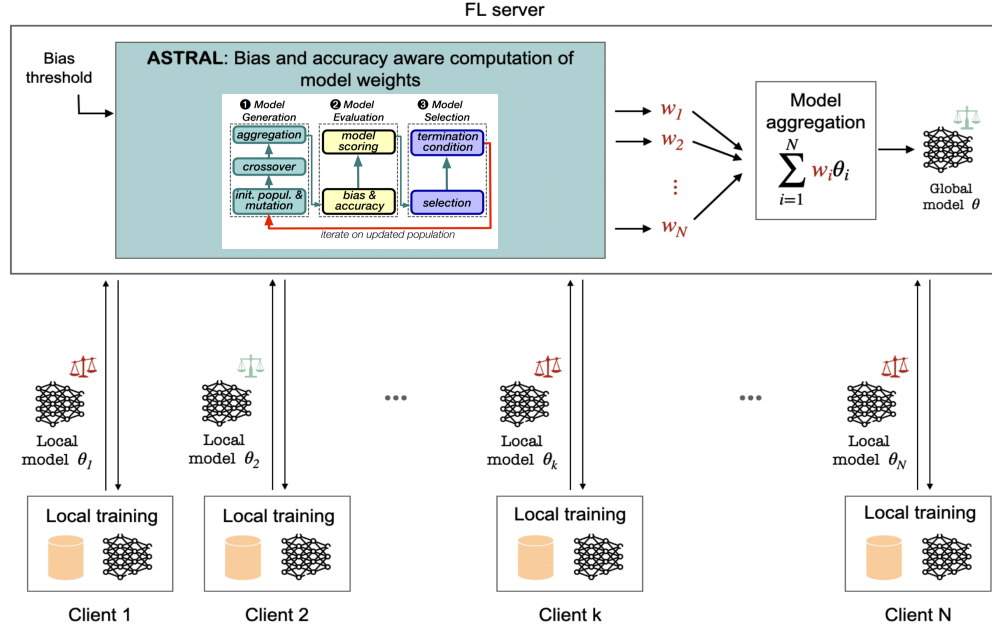


Fig. 4. Overview of ASTRAL architecture

ASTRAL's pipeline at the server level consists of an iterative optimization process with three essential steps to explore diverse aggregation weights and select the best among them with regard to our objectives, as shown in Fig. 5. Initially, the server receives the local models of FL clients and the bias threshold as input and launches the ASTRAL's pipeline. The first step in each iteration of the pipeline involves Model Generation (Fig. 5-❶) which uses the Differential Evolution algorithm (DE) to construct a population of potential aggregation weights and corresponding global models [61]. The second step, *i.e.*, Model Evaluation (Fig. 5-❷), targets the assessment of the accuracy and the bias of each global model created in the first step and computes upon those two measurements each global model's score. Finally, Model Selection (Fig. 5-❸) uses the scores computed in Fig. 5-❷ to select the best global models to keep in the population. The process iterates and loops back to Fig. 5-❶ with the updated population, until reaching a predefined termination condition. It then outputs the final model which is the model that has the best score of all the population with regard to the objectives of the constrained optimization problem (see §4.1). Once the iterative process stops, the current FL round is terminated, and the selected model will be sent to the FL clients at the beginning of the next round.

4.3 ASTRAL Design Principles

Here, we describe in details the steps of the ASTRAL pipeline, presented in Algorithm 1. The pipeline starts by the random initialization of a first population of potential candidates for aggregation weights $\bar{w}$. ASTRAL iteratively modifies this population through the three introduced processes: Model generation (lines 7-11 of algorithm 1), Model evaluation (line 12 of algorithm 1) and Model selection (lines 13-16 of algorithm 1). At the end of the pipeline, the best aggregation weight is selected and applied on the clients models to output the optimized global model.

4.3.1 Model Generation. During model generation, current aggregation weights candidates are modified to create new candidate solutions called offspring candidate solutions. Precisely, if P is the current population of candidate solutions, then this module creates for each $\bar{w}_i \in P$ an offspring individual $T_{\bar{w}_i}$ using the *mutation* and *crossover* operators from the DE algorithm, as follows. First, the mutation operator selects randomly two members of the population e_1 and e_2 . Their difference, scaled by a factor F , is used to mutate the best member of the population b , as follows: $m_{\bar{w}_i} = b + F * (e_1 - e_2)$ (line 8). Then, the crossover operator creates the offspring $T_{\bar{w}_i}$ by randomly selecting a component w_k from $\bar{w}_i$'s components, i.e., $\{w_1 \dots w_N\}$, and replacing it with the corresponding component from the mutant vector $m_{\bar{w}_i}$ (line 9). The mutation and crossover operators introduce randomness and maintains diversity of populations. Finally, the weight candidate solutions and their offsprings are used to compute global aggregated models (lines 10-11).

4.3.2 Model Evaluation. This step evaluates the accuracy and bias of the models generated in the Model Generation part, and assigns a score to each using a fitness function (line 12). To define an appropriate fitness function for our problem, we follow the known approach of black-box constrained optimization where the fitness function is composed of two main terms: the objective function, and a global penalty term [33, 44]. The objective function term represents the goal of the problem and measures how well a solution performs in achieving this goal, which in our case is model's accuracy. The global penalty term is to be subtracted from the objective function in the fitness function formula. Its role is to take into account all the constraints imposed by the problem by penalizing solutions that do not respect the constraints, which in our case are the bias constraints. The global penalty term is the sum of penalties of the different constraints of the problem. A penalty is usually proportional to the deviation from the constraint and takes various forms, such as squared penalty and absolute penalty. Squared penalty gives more penalty to solutions that deviate more from the constraint. When the deviation from the constraint is below 1, the squared penalty gives this deviation even smaller weight. In our case, the deviation from the bias constraints is generally below 1. We apply on our constraints the absolute penalty that does not exacerbate or

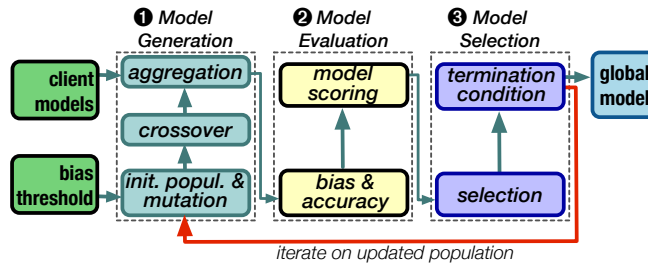


Fig. 5. The ASTRAL pipeline at the FL server-side

Algorithm 1: ASTRAL bias mitigation pipeline at server level

Inputs : Θ : models' parameters of FL clients
 ϵ : Global FL model bias threshold
Output: $\theta_{w_{best}}$: Global FL model's parameters meeting ϵ bias threshold with the highest model accuracy

```

1 Initialise population  $P$  with random vectors of weights of FL clients' updates
2 Initialise  $max\_iter$  to the maximum #iterations of Differential Evolution
3 Initialise  $nb\_iter$  to 0,  $P_{new}$  as empty, and  $termination\_condition$  to False
4 do
5   foreach vector of weights of FL clients' updates  $\overline{w}_i \in P$  do
6     // Model Generation
7     Select randomly  $e_1$  and  $e_2$  from  $P$ 
8     Select the best individual  $b$  from  $P$ 
9     Create mutant vector  $m_{\overline{w}_i}$  using  $e_1, e_2, b$ 
10    Create offspring vector  $T_{\overline{w}_i}$  using DE crossover operator between  $\overline{w}_i$  and  $m_{\overline{w}_i}$ 
11    Create model  $\theta_{\overline{w}_i}$  by applying weights  $\overline{w}_i$  on clients' model parameters  $\Theta$ 
12    Create model  $\theta_{T_{\overline{w}_i}}$  by applying weights  $T_{\overline{w}_i}$  on clients' model parameters  $\Theta$ 
13    // Model Evaluation
14    Compute score of  $\theta_{\overline{w}_i}$  and score of  $\theta_{T_{\overline{w}_i}}$  with regard to  $\epsilon$  bias threshold and high accuracy, and store them in  $Score(\theta_{\overline{w}_i})$ 
15    and  $Score(\theta_{T_{\overline{w}_i}})$  respectively
16    // Model Selection
17    if  $Score(\theta_{T_{\overline{w}_i}}) > Score(\theta_{\overline{w}_i})$  then
18      | Add  $T_{\overline{w}_i}$  to  $P_{new}$ 
19    else
20      | Add  $\overline{w}_i$  to  $P_{new}$ 
21   $nb\_iter++$ 
22  Replace the content of  $P$  by the content of  $P_{new}$ 
23  if all elements of  $P$  have similar scores, or if  $nb\_iter = max\_iter$  then
24    | Set  $termination\_condition$  to True
25 while  $termination\_condition \neq True$ ;
26 Find among elements of  $P$  the vector of weights  $w_{best}$  with the highest score
27 Compute new global FL model  $\theta_{w_{best}}$  by applying  $w_{best}$  on  $\Theta$ 
28 return  $\theta_{w_{best}}$ 

```

underestimate deviations from the constraints. Thus, the fitness function of ASTRAL is defined as follows:

$$Score(\theta_{\overline{w}}, \epsilon) = A(\theta_{\overline{w}}) - \sum_{|\beta_{S_j}(\theta_{\overline{w}})| > \epsilon} |\beta_{S_j}(\theta_{\overline{w}})| \quad (9)$$

Where $Score(\theta_{\overline{w}}, \epsilon)$ is the fitness of a solution candidate $\overline{w}$, with ϵ being the predefined bias threshold, $\theta_{\overline{w}} = \overline{w} \cdot \Theta$ being the corresponding global model to $\overline{w}$, $A(\theta_{\overline{w}})$ its accuracy and $\beta_{S_j}(\theta_{\overline{w}})$ the model bias with regard to the sensitive attribute $S_j \in \{S_1 \dots S_{sa}\}$. The global penalty term is computed by summing bias penalties of all the sensitive attributes S_j whose bias exceeds the predefined bias threshold ϵ . This fitness function is then used to evaluate the fitness scores of candidate models during the model evaluation step. The computation of these fitness scores and global metrics in practice is described afterwards in the dedicated paragraph.

4.3.3 Model Selection. After the models' evaluation, the system uses the *selection* operator to compare the parent $\bar{w}$ with its offspring $T_{\bar{w}}$ and only keeps the best in terms of model fitness in the new population called P_{new} (line 13-16). After iterating through all current elements in P , the latter is replaced with the newly updated population P_{new} . Finally, the system has a termination condition to determine whether it should finish ASTRAL execution for the current round or send back the updated population for a new iteration (lines 20-21). This termination condition can be whether the number of iterations done through ASTRAL has reached a maximum value denoted max_iter , or if all elements of P have similar scores. If the termination condition is respected, then ASTRAL identifies the best individual w_{best} , which is the aggregation weight that results in the model with the best score (line 21). It then computes the associated aggregated model as the new global model $\theta_{w_{best}}$ (line 24) for the current round. The new global model is sent back to the clients to use at the beginning of the next FL round.

The described ASTRAL's pipeline is executed at each FL round. It may occur that for some rounds, the final population does not contain solutions that conform to the bias constraints. In that case, the selection operator selects solutions that do not respect the bias constraints. To overcome this issue, we rely on the exploration capabilities of ASTRAL, which we explain in the next section and which showed high effectiveness in finding high quality solutions during the evaluation process (see §5). Another efficient strategy we employ is increasing when necessary the size of the populations to be explored in order to explore better the search space. Nevertheless, our empirical analysis shows that such scenario is infrequent. Our approach demonstrated its ability in generating high-quality feasible solutions, in the FL early rounds (see §5). At the end of the FL rounds, ASTRAL outputs the best global model among the rounds' global models in terms of bias and accuracy objectives.

4.3.4 How FL Server Calculates Global Metrics. A pivotal factor in mitigating bias within the ASTRAL framework is the server's ability to compute the global model accuracy and bias metrics in each iteration of ASTRAL's pipeline (See Eq. 9). The server uses the exchanged model updates and a proxy dataset at its possession to get the required metrics, obviating the need of sharing additional sensitive information. Using proxy dataset has found application in previous contexts such as [40] and [6]. In these works the proxy dataset is used for knowledge distillation enabling the transfer of knowledge from individual client models to the global model. Likewise, FedLAW uses a globally consistent proxy dataset and the gradient descent algorithm to optimize FL aggregation [39]. Within ASTRAL, the use of this proxy dataset gives a strategic advantage in mitigating bias. In §5.12, we show the effectiveness of our proposed method, even when the proxy dataset diverges from the training data distribution.

To evaluate global metrics on the server-side without requiring a proxy dataset, alternative methods exist. One method involves cryptographic techniques such as secure aggregation [20], where global metrics are computed by aggregating values from participating FL clients at the server without revealing individual values. However, this approach introduces a computational overhead. Another approach, similar to [62], involves considering a subset of the clients as proxy data holders and computing the required metrics from these clients, while not including them in the training. Additionally, in the absence of proxy data, artificial data can be generated using existing privacy-preserving data generation methods in FL [73].

4.4 Analytical Insights

4.4.1 Maximizing FL Model Accuracy. ASTRAL finds the solution that maximizes accuracy by integrating the accuracy in the fitness function. This allows to assign high fitness scores to high accuracy solutions leading to their systematic selection as the best solution candidates by our pipeline. Furthermore, ASTRAL is able to ensure globally optimal solutions that maximize accuracy as much as possible by optimizing the exploration of the search space. First, ASTRAL leverages mutation, crossover, and selection operations, which are critical in forming sequences of populations and generating both diverse and good quality solutions. These operations are applied iteratively at the server level, enabling ASTRAL to explore the search space more effectively. Then, ASTRAL runs

through several FL rounds, which further enhances exploration, as explained in the following. At the start of each round, the pipeline of ASTRAL sends the global model to the clients for training. The clients subsequently submit their models (locally trained) to ASTRAL at the end of the round. Each client has its own local model, which represents its unique knowledge learned from its local data. The pipeline of ASTRAL then explores new populations of aggregation weights to apply to these local models. The exploration of new aggregation weights leads to the exploration of new global models. The explored global models of each FL round are formed via a linear combination of the local models' parameters using the aggregation weights candidates. Formally, at each round, ASTRAL explores a potentially new subspace of the global model search space, equivalent to the span generated from the algebraic basis formed by the local models of the clients of the round. The span generated by the algebraic basis formed by the local models of the clients for a round t is formally defined by the following equation:

$$\text{Span}_t(\theta_1^t, \theta_2^t, \dots, \theta_N^t) = \{w_1.\theta_1^t + w_2.\theta_2^t + \dots + w_N.\theta_N^t \mid w_1, w_2, \dots, w_N \in \mathbb{R}\} \quad (10)$$

where $\theta_1^t, \theta_2^t, \dots, \theta_N^t$ are the local models of the clients at round t . By combining the local models of clients in different FL rounds, ASTRAL can leverage the collective knowledge and learning from various training perspectives, depending on the FL round. The exploration of different spans generated by the local models as we progress through multiple FL rounds enhances the exploration of the search space.

4.4.2 Ensuring FL Model Bias Below a Threshold. In ASTRAL, the bias constraints are enforced by including them in the penalty term of the fitness function defined in Eq. 9. Specifically, the penalty assigned to a model given a bias objective is computed as the sum of all the left terms of bias inequality constraints that are not respected, as follows:

$$\text{penalty}(\theta_w, \epsilon) = \sum_{S_j \in [S_i, S_{sa}]} \max(0, \beta_{S_j}(\theta_w) - \epsilon) \quad (11)$$

This penalty ensures that models that exceed the bias threshold are penalized proportionally to the magnitude of their bias, while the models that do not exceed the bias threshold are not penalized and thus are the ones selected as potential solutions. The exploration capacities of ASTRAL described above have also crucial role at ensuring the bias objective. By leveraging the DE operators, and by exploring different spans generated by the local models of different clients in each FL round, ASTRAL effectively explores diverse subspaces of the global model search space. This allows to discover a diverse array of solutions that meet the bias constraints.

4.4.3 Generic Approach to Handle Several Bias Metrics. ASTRAL's generic framework can accommodate a wide range of bias metrics. Its black-box optimization approach does not require any specific conditions on the function being optimized, or the constraints, such as continuity or derivability. Moreover, leveraging black-box optimization offers another advantage to ASTRAL compared to existing methods. ASTRAL stands out as a bias mitigation solution by successfully optimizing the objective bias constraint itself without relying on approximations or surrogate functions. For instance, AgnosticFair [16] adopts an exact optimization technique and thus resort to approximate the statistical parity difference metric with its convex proxy, the decision boundary fairness metric [78]. However, this approximation fails to provide guarantees regarding the mitigation of statistical parity difference [72, 78]. In contrast, ASTRAL ensures that the constraints in terms of the real objective are respected, avoiding the potential pitfalls associated with inaccurate approximations.

4.4.4 Algorithm Complexity Analysis. The algorithm complexity of the proposed solution outlined in algorithm 1 depends on the algorithm complexity of ASTRAL's pipeline steps. The pipeline functions over a maximum of $\max_{iter}$ iterations. In each iteration, the aggregation weights population of size $|P|$ is browsed, and the model generation, evaluation and selection operations are applied. The model generation process (lines 6-11) comprises the selection of the best solution candidate, which cost depends on $|P|$. Then the mutation and crossover operations are applied, with costs proportional to number of the participating clients k that represents the dimension of

the aggregation weights. The model generation process also includes the cost of creating the global models corresponding to the mutant and crossover vectors, dependant on $k \cdot d$, where d is the number of parameters of the learning model. The model evaluation phase (line 12) cost depends on c , the cost of applying the fitness function on a candidate solution. The fitness function computations costs depend on how the FL server calculates global metrics (See §4.3). Finally, during the model selection phase (lines 13-16), the population is updated based on the evaluation scores, introducing a constant cost. When the stopping criteria is met, in the worst case attaining the predefined maximum iteration count max_{iter} , the weights with the highest scores are selected, and used to create the new global FL model (lines 23-24). This operation entails a cost depending on $|P| + (k \cdot d)$. Thus, the overall complexity is formulated as follows:

$$C = O(max_{iter} \cdot |P| \cdot (|P| + k + k \cdot d + c) + |P| + k \cdot d) \quad (12)$$

As presented in previous studies [24], population size does not exceed 50 for most engineering problems. Thus, model generation operations are less intensive than the model evaluation and $|P|$ is lower than c . The algorithm complexity of the proposed method is expressed as Eq. 13.

$$O(max_{iter} \cdot |P| \cdot (k \cdot d + c)) \quad (13)$$

Taking this complexity into consideration helps to determine the various factors impacting the computational cost of ASTRAL. Reducing both of population size and the maximum iteration count and adopting low-cost techniques to get models' metrics reduces significantly ASTRAL's cost, and allows it to scale efficiently to complex models and data, and high number of clients. The calibration of these parameters depends on the unique requirements of the given scenario and the characteristics of the used data and models.

5 EXPERIMENTAL EVALUATION

We assess the effectiveness of our bias mitigation proposal with three popular bias metrics, seven widely used datasets and various experimental setups. Specifically, we seek to answer the following questions:

- How efficient is ASTRAL in reaching bias constraint compared to other techniques? (in §5.4)
- How efficient is ASTRAL in maximizing accuracy while reaching bias constraint compared to other techniques? (in §5.5)
- In what ways might the mitigation of a single sensitive attribute result in more bias? (in §5.6)
- How well does ASTRAL meet different fairness objectives, from the least stringent to the most stringent? (in §5.7)
- How scalable is ASTRAL with regard to the data size? (in §5.8)
- How scalable is ASTRAL with regard to the number of clients? (in §5.9)
- How scalable is ASTRAL with regard to different FL clients participation ratios? (in §5.10)
- How robust is ASTRAL with regard to data heterogeneity? (in §5.11)
- What is the impact of the data distribution of the proxy dataset on ASTRAL's performance? (in §5.12)
- How stable is ASTRAL's bias mitigation? (in §5.13)
- What is the cost of ASTRAL in terms of bias, accuracy, and communication efficiency trade-offs? (in §5.14)

We first describe the datasets, implementation details and experimental setup, and the bias mitigation techniques against which we compare. Then, we present the empirical evaluation of ASTRAL.

5.1 Datasets and Models

We use seven real-world datasets, chosen specifically for their well-known bias issues. Table 4 summarizes their characteristics.

Table 4. Characteristics of the real-world datasets used in our evaluation. We include: topic, the geographical location (Loc.), the total number of raw and cleaned records, the number of available attributes (#Attr.), the sensitive attributes and the target class.

Dataset	Topic	Loc.	#Records (cleaned)	#Attr.	Sensitive Attributes	Other Attributes	Target Class
ARS	Healthcare	ASTL	75,128 (75,128)	9	<i>gender</i>	acceleration on axis x,y,z, signal strength	activity
MobiAct	Healthcare	GR	16,756,325 (1,852,861)	16	<i>age, gender</i>	activity, roll, pitch, acc_x	activity
CelebA	Emotion recognition	CN	202,599 (202,599)	3074	<i>age, gender</i>	Images' pixels values	emotion
KDD	Finance	US	299,285 (272,507)	41	<i>age, gender, race</i>	education level, work category, native country	income
DC	Finance	NL	60,420 (60,420)	12	<i>age, gender</i>	citizenship, birth country education level	job
Adult	Finance	US	48,842 (48,842)	15	<i>age, gender, race</i>	education level, hours worked/week, native country	income
MEPS	Healthcare	US	35,428 (33,401)	134	<i>gender, race</i>	occupation, US region, income	health

ARS. The ARS ubiquitous dataset is used for activity recognition of healthy older people using a batteryless wearable sensor dataset [59]. It has been collected in two clinical rooms equipped with four and three RFID reader antennas respectively. Fourteen volunteers aged between 66 and 86 years were trialed. Each wore a wearable sensor (W2-ISP) at the sternum level, and undertook a series of scripted activities including (i) lying on a bed, (ii) sitting on a bed, (iii) sitting on a chair, and (iv) ambulating. The classification task is to predict whether a person is lying or not for ambulatory monitoring. The only sensitive attribute present in this dataset is gender. To accomplish the learning task of ARS, we use a linear support vector machine model. In the case of FCFL which does not support SVM models, we employ logistic regression.

MobiAct. The MobiAct dataset has been designed for smartphone-based human activity recognition [70]. It contains data collected from accelerometers, gyroscopes, and orientation sensors of smartphones. It captures various daily activities performed by 66 participants with more than 3200 trials. Our primary objective is to identify individuals who are standing and those who are not. The sensitive attributes are gender and age. To accomplish the learning task, we employ a 3-layer Multilayer Perceptron (MLP) model [51].

CelebA. CelebA is a public dataset containing over 200,000 facial images of 10,177 different celebrities [41]. Each image is annotated with 40 attribute labels including gender and age as sensitive attributes. As learning task we consider emotion recognition, more precisely smile detection. We use a deep learning model composed of the ResNet18 convolutional neural network [26] and a fully connected neural network. ResNet18 is used to process the dataset images' pixels while the other network is used to process ResNet18's outcomes and the sensitive attributes of the images. The whole model outputs the classification result.

KDD. The KDD dataset contains weighted census data extracted from the 1994 and 1995 population surveys conducted in the US [17]. It includes 299,285 records with 41 attributes including age, education level, race, gender, *etc.* The prediction task is to determine whether an individual's income exceeds \$50K or not. To perform this task we use a logistic regression model. Sensitive attributes are *age*, *gender* (men, women), and *race*. We binarized the *race* attribute into two groups, considering as *privileged* the Whites, Asians and Pacific Islanders, and *unprivileged*

all others. Similarly, we binarized the *age* attribute by considering *privileged* all people aged between 30 and 60, and *unprivileged* all the others.

DC. The Dutch Census dataset collects census information in the Netherlands in 2001 [69]. It includes 60,420 records with 12 attributes. The prediction task is to classify a person's occupation as having high or low prestige, the model used for this task is logistic regression. Sensitive attributes are *age* and *gender*. We binarized the *age* by considering *privileged* those whose *age* is below 50, and *unprivileged* the others.

Adult. The Adult dataset contains 48,842 records and a total of 15 attributes [36], extracted from the 1994 US Census database. It includes individuals' information such as age, education level, race, gender, *etc.* The prediction task is the same as KDD, considering the same sensitive attributes. The used model here is logistic regression as well. We applied the same binarization process to the *race* and *age* attributes of Adult as in KDD.

MEPS. The MEPS dataset originates from the Medical Expenditure Panel Survey in the US [10]. It consists of 35,428 records with 134 attributes. The task is to predict whether an individual has used medical facilities more than 10 times. To do so, we apply a logistic regression model. Sensitive attributes are *gender* and *race*. We binarized the *race* attribute, considering Non-Hispanic Whites as *privileged*, and the others as *unprivileged*.

For all the datasets, the binarization of non-binary sensitive attributes is done through a preprocessing step. The binarization is necessary since bias metrics are defined for binary sensitive attributes only. It involves clustering the sensitive attribute values into two groups based on the positive outcome ratio computed for each group using the ground truth. The group with higher ratios is classified as privileged, while the other group is unprivileged. The value of the sensitive attribute is then replaced by the obtained binary representation. Additionally, in each group of experiments performed with the same dataset, we initialize the training with the same initial model. It provides a fair ground for experimentation and comparison with baselines.

5.2 Implementation and Experimental Setup

ASTRAL is implemented in Python (v3.9) and PyTorch (v1.10). We use the SciPy implementation of Differential Evolution [58]. By default, we set in our experimentations the initial population size to 15, we set 8 parallel workers, and keep the other method parameters equal their default values. We run our experiments on a testbed platform with the following characteristics: 2× Intel Xeon E5-2650 v4 (14 cores), 128 GB of RAM and 2× NVIDIA GTX 1080 Ti. We simulate a network with a download speed of 20 Mbps and an upload speed of 5 Mbps. Datasets are split into three distinct subsets, namely training set (80%), test set (10%), and proxy dataset (10%) to compute global metrics (§4.3). To emulate FL clients and run the FL process, the training set is distributed according to the FL settings of each experiment. In our experiments, data is distributed among FL clients in a non-IID setting using Dirichlet function with regard to a particular sensitive attribute for each dataset [29]. This means that each FL client receives a different distribution of data based on the considered sensitive attribute. For each value v^j of the sensitive attribute S_j , we sample $p_{v^j} \sim \text{Dir}(\alpha)$ with α parameter, and allocate a portion $p_{v^j, k}$ of the data points with $S_j = v^j$ to client k . The heterogeneity of the distributions across clients is controlled via α parameter. When $\alpha \rightarrow \infty$, the distributions are IID, whereas smaller values of α lead to more heterogeneous distributions. In our experiments, we generate clients data distributions with $\alpha = 0.02$. The FL number of clients varies depending on the dataset, with respectively 4 clients for MEPS and CelebA, 5 for KDD, 10 clients for both ARS, DC and Adult. For MobiAct, we use a subset of the whole dataset (12 clients) that exhibits bias since the dataset as a whole, with a total of 66 clients, does not demonstrate bias. Logistic regression models are applied to Adult, MEPS, KDD and DC datasets. For Adult and MEPS datasets, we use a batch size of 128 and a learning rate of 0.01, while for KDD, we use a model with the same batch size and a learning rate of 0.001. In the case of DC, we use a batch size of 256 and a learning rate of 0.0005. For ARS dataset, we use a batch size of 1024 and a learning rate of 0.001 with SVM model. For MobiAct, we use a Multilayer perceptron with a batch size of 256 and a learning rate of 0.01. For CelebA, we use a ResNet18 based model with a batch size of 1024 and a dynamic learning rate initialized to 0.01.

In several bias mitigation works [11, 16], the typical bias threshold is considered to be 3%. Thus, we consider this threshold per default in our study. Further details of the implementation and experimental setup can be found in ASTRAL’s repository, where we provide the software prototype, the used datasets, and the configuration settings: <https://github.com/FL-Bias/ASTRAL>

5.3 Baselines

We compare ASTRAL against the following FL methods.

FCFL [11]. This state-of-the-art FL bias mitigation method formulates a per-client bias mitigation problem, where the objective is to guarantee a bias constraint on each client local distribution while maintaining similar model accuracy across all clients. This is done through gradient-based optimization. We use the original FCFL implementation [12], based on PyTorch 1.6 and Python 3.7.

FairFed*. As far as we know, the software prototype of FairFed [20] is not publicly available. We implement the algorithm presented in FairFed, and adapt it to reweigh client’s updates based on their bias and the predefined bias threshold. More precisely, given a sensitive attribute S_j and a bias metric β , in order to mitigate bias in FairFed*, a factor is applied to the classical FL weight w_k from Eq. 1, depending on the amount of bias $\beta_S(\theta_k)$ of client k . Formally, we compute the weights w'_k as $w'_k = w_k \cdot g(\beta_S(\theta_k))$, where:

$$g(\beta_S(\theta_k)) = e^{-b(-\log(1-|\beta_{S_j}(\theta_k)|))^a} \quad (14)$$

where a and b are two constant parameters. We tune them in each experiment to obtain the best model with regard to accuracy and bias objectives. The function g is inspired by the Prelec reweighing function [55]. Thus, FairFed* favors clients with lower model bias.

FairFL*. FairFL proposes a client selection policy that relies on a reinforcement learning algorithm and selects participants depending on their contribution to the final model’s bias [80]. Since the original software prototype of FairFL is not publicly available, we implemented a simple heuristic, referred to as FairFL*, that selects at each round clients whose local models’ bias is below the given threshold ϵ . More formally, w_k in Eq. 1 is set to 0 for every θ_k for which there exists a sensitive attribute $S_j \in \{S_1 \dots S_{s_a}\}$ such that $\beta_{S_j}(\theta_k) > \epsilon$.

FedAvg [45]. We also consider the case of FedAvg, where no bias mitigation is applied, and where default FL model aggregation and client selection are used (*cf.* §2.1). This allows us to compare the FL model accuracy obtained in ASTRAL with the case where no bias mitigation is applied.

We note that among the mentioned techniques, only FairFL* supports the existence of multiple sensitive attributes in data. Instead, FCFL and FairFed* can only handle single sensitive attribute bias mitigation. Additionally, it is not a trivial task to adapt the two latter methods to handle multiple sensitive attributes bias mitigation.

5.4 Comparison of ASTRAL against Existing FL Bias Mitigation Mechanisms

We start by evaluating how ASTRAL addresses bias when considering several sensitive attributes simultaneously. We consider multiple sensitive attributes depending on the dataset columns as follows: for KDD and Adult, we consider gender, race, and age; for DC, MobiAct and CelebA, sensitive attributes are gender and age; for MEPS sensitive attributes are gender and race. The ARS dataset contains only one sensitive attribute (gender), but we still include the results of applying ASTRAL on it. To assess its effectiveness, we compare ASTRAL against FedAvg and FairFL*. We only use FairFL* because it is the only method that can handle multiple sensitive attributes, as mentioned in §5.3. In all experiments, we fixed a bias constraint of 3%, and our evaluation includes SPD, EOD, and DI metrics. We report the model’s bias results in Fig. 6. We observe that ASTRAL matches the bias constraint for all the multiple sensitive attributes scenarios with the different datasets and bias metrics. FedAvg and FairFL* do not satisfy the bias constraint. We explain this as follows. First, FedAvg does not mitigate bias by design, so it yields biased models. Then, for FairFL*, the method takes a highly pessimistic approach. In most of the experiments,

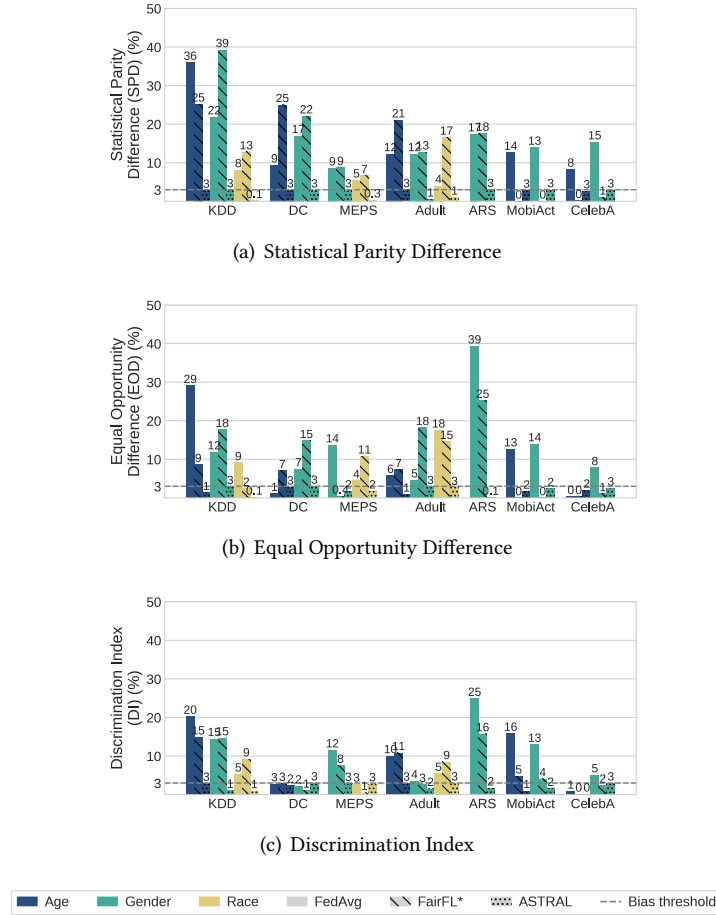


Fig. 6. Bias mitigation with regard to multiple sensitive attributes using SPD, EOD and DI metrics – KDD, DC, MEPS, Adult and ARS datasets – With respectively 5, 10, 4, 10 and 10 FL clients and a bias threshold of 3%.

it does not select any client model for aggregation from the first round because their local models are already biased; so it just outputs the initial model. Exceptional cases are when FairFL* is applied on DC considering DI metric, on MobiAct considering SPD and EOD, and on CelebA considering the different bias metrics. In the latter cases, FairFL* is able to select clients that meet the bias constraint, resulting in an unbiased model. However, since FairFL* allows less clients to join the training, it achieves low accuracies compared to ASTRAL. For example, for DC when considering DI, FairFL* achieves 81% while ASTRAL achieves 83%. More bias and accuracy trade-offs results are shown in §5.5.

5.5 Trade-off Between Bias and Model Accuracy

In the following, we evaluate the effectiveness of ASTRAL in terms of bias and accuracy trade-offs. To do so, we report measurements of bias with regard to multiple sensitive attributes along with measurements of accuracy. We consider the baseline system without bias mitigation (*i.e.*, FedAvg), FairFL*, and ASTRAL. The

measurements are collected for each method and dataset at 10 representative rounds. The representative rounds are the percentiles 10^{th} , 20^{th} ... until the 100^{th} of the FL rounds' ordered values. In addition to these rounds, we include the performance of each method at the round where it achieves its highest accuracy while maintaining the bias constraint. Due to space limitations, we present the results considering different bias metrics for ARS (Fig. 7(a)), MobiAct (Fig. 7(b) and Fig. 7(c)), CelebA (Fig. 7(d) and Fig. 7(e)), and Adult (Fig. 7(f), Fig. 7(g), and Fig. 7(h)). The bias threshold is fixed to 3%.

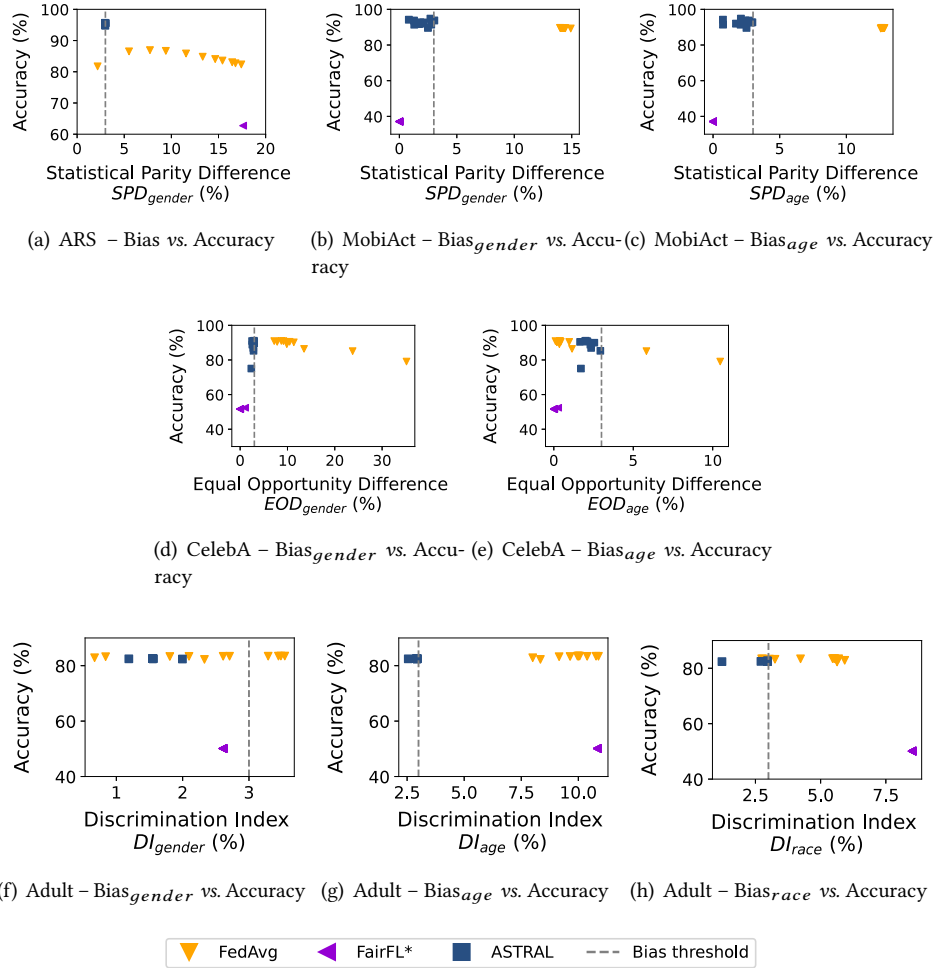


Fig. 7. Trade-off between bias and model accuracy

The reported measurements indicate that ASTRAL consistently maintains the bias objective for the different sensitive attributes of the considered datasets across the considered rounds. Simultaneously, ASTRAL achieves higher model accuracies compared to the other bias mitigation technique FairFL* and this for all the datasets, we refer this to FairFL*'s clients selection approach that results in low accuracies. ASTRAL is even able to reach

higher accuracy compared to the baseline which does not mitigate bias by design in the case of ARS and MobiAct. This result is explained by the effectiveness of ASTRAL's algorithm in selecting the best aggregation weights that maximize the system accuracy, allowing to reach the best trade-offs of bias and accuracy in our experiments.

5.6 Mitigating Single Sensitive Attribute Bias May Induce a Higher Bias With Regard to the Other Sensitive Attributes

To measure how bias mitigation on a single sensitive attribute affects the model bias on other sensitive attributes within the same dataset, we compare ASTRAL with the two other state of the art methods, that handle a single sensitive attribute, namely FCFL and FairFed*. We fix a bias constraint of 3%, and we consider one single sensitive attribute for each dataset: *age* for KDD and Adult, *race* for MEPS and *gender* for DC, ARS and MobiAct. Table 5(a), Table 5(b) and Table 5(c) show the results respectively for a set of experiments done on the SPD, EOD and DI bias metrics. The second column from left for each table reports bias measurements on the sensitive attribute considered during bias mitigation, while the two last columns report bias measurements on the remaining sensitive attributes.

We observe that focusing solely on mitigating bias with respect to one sensitive attribute may have unwanted consequences with regard to the others. For instance, the EOD of FairFed* for KDD dataset on the age sensitive attribute, shown in Table 5(c), is equal to 3%, showing that the model bias on the age sensitive attribute is mitigated. However we see that for the same scenario, the EOD on the race sensitive attribute increased when comparing FedAvg and FairFed*, from 12% to 21%. So, mitigating the bias towards one sensitive attribute does not guarantee mitigating the bias for the other sensitive attributes, and worse, it may degrades them. The same

Table 5. Exacerbation of bias when mitigating considering a single sensitive attribute, with SPD, EOD, and DI – the first bias column from left represents the bias with regard to the sensitive attribute considered by the mitigation mechanism

(a) Mitigating bias with regard to SPD

Adult			
System	SPD _{age} (%)	SPD _{race} (%)	SPD _{gender} (%)
FedAvg	13	4	13
FCFL	1	9	19
ASTRAL	3	1	3
MEPS			
System	SPD _{race} (%)	SPD _{gender} (%)	
FedAvg	5	8	
FCFL	2	10	
ASTRAL	0	3	
MobiAct			
System	SPD _{gender} (%)	SPD _{age} (%)	
FedAvg	13	14	
FairFed*	3	21	
ASTRAL	3	3	

(b) Mitigating bias with regard to DI

KDD			
System	DI _{age} (%)	DI _{race} (%)	DI _{gender} (%)
FedAvg	20	5	14
FairFed*	0	22	8
ASTRAL	3	1	1
Adult			
System	DI _{age} (%)	DI _{race} (%)	DI _{gender} (%)
FedAvg	10	6	4
FairFed*	0	13	13
ASTRAL	3	3	2

(c) Mitigating bias with regard to EOD

KDD			
System	EOD _{age} (%)	EOD _{gender} (%)	EOD _{race} (%)
FedAvg	29	9	12
FairFed*	3	2	21
ASTRAL	1	3	0
DC			
System	EOD _{gender} (%)	EOD _{age} (%)	
FedAvg	8	4	
FairFed*	3	6	
ASTRAL	3	3	
Adult			
System	EOD _{age} (%)	EOD _{gender} (%)	EOD _{race} (%)
FedAvg	8	8	17
FairFed*	0	10	20
ASTRAL	1	3	3
MEPS			
System	EOD _{race} (%)	EOD _{gender} (%)	
FedAvg	6	14	
FairFed*	2	16	
ASTRAL	2	2	

phenomenon is observed for SPD and DI with both FCFL and FairFed* for Adult, MEPS, KDD and MobiAct as shown in Table 5(a), Table 5(b), and Table 5(c). On the contrary, ASTRAL is able to mitigate bias with regard to all sensitive attributes simultaneously.

5.7 Evaluation of Bias Mitigation under Different Bias Constraints

To determine if ASTRAL is able to enforce different stringent bias constraints on the FL model, we run our solution under more or less strict bias constraints, using KDD dataset with age, gender and race as sensitive attributes, and considering the SPD metric. The results are presented in Fig. 8. It shows the global model's bias with regard to the three sensitive attributes, measured in different setups with different bias limits imposed. We observe that ASTRAL is able to successfully provide FL a model that satisfies the objective for all the sensitive attributes, even when hardening the bias constraint.

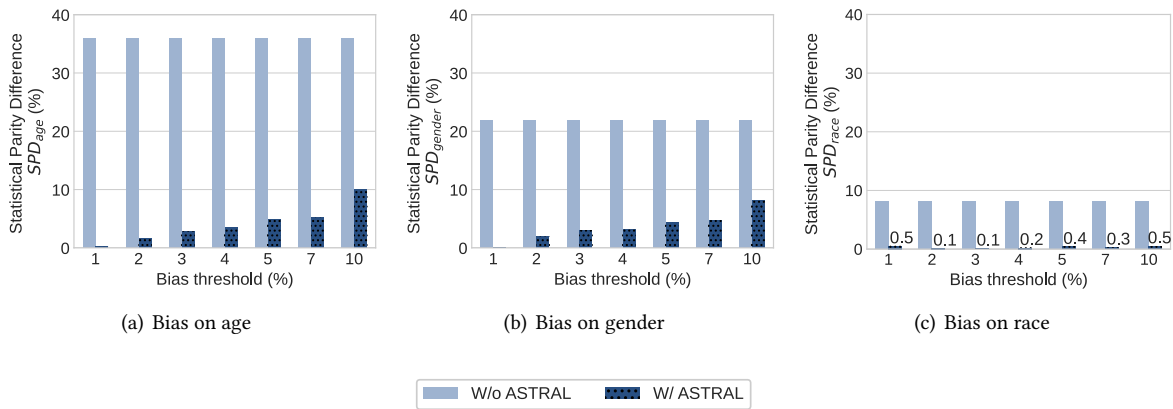


Fig. 8. Effectiveness of bias mitigation under different bias thresholds

5.8 FL System Scalability with Regard to the Data Size

We analyze how ASTRAL performs when varying the size of the training data. For this set of scalability experiments, we consider the SPD as the bias metric. We also consider the KDD dataset, split in several partitions of data, each containing a different percentage of data records of the whole dataset. We include the data from each partition in the ones with a larger number of records. Namely, the 10% partition is included in the 20% partition, and so on, until reaching the 100% partition, which contains the whole KDD train set. Fig. 9 reports bias measurements without bias mitigation (*i.e.*, FedAvg) and with ASTRAL, and shows how ASTRAL performs as a function of the partition size for KDD. We observe that ASTRAL's trend is extremely stable for this dataset. ASTRAL maintains the bias metric to 3%, ensuring the bias objective as data size increases, and this for the three sensitive attributes age, gender and race, as shown by Fig. 9(a), Fig. 9(b), and Fig. 9(c). In summary, ASTRAL's bias mitigation is not affected by the data scale.

5.9 FL System Scalability with Regard to the Number of Clients

We analyze how ASTRAL performs with the number of FL client. We consider the KDD dataset, where we configure FL setups ranging from 10 clients to 100 clients. We include the data from a specific client setup in the data of the other FL setup with a more significant number of clients. Namely, the 10-client setup data is included in the

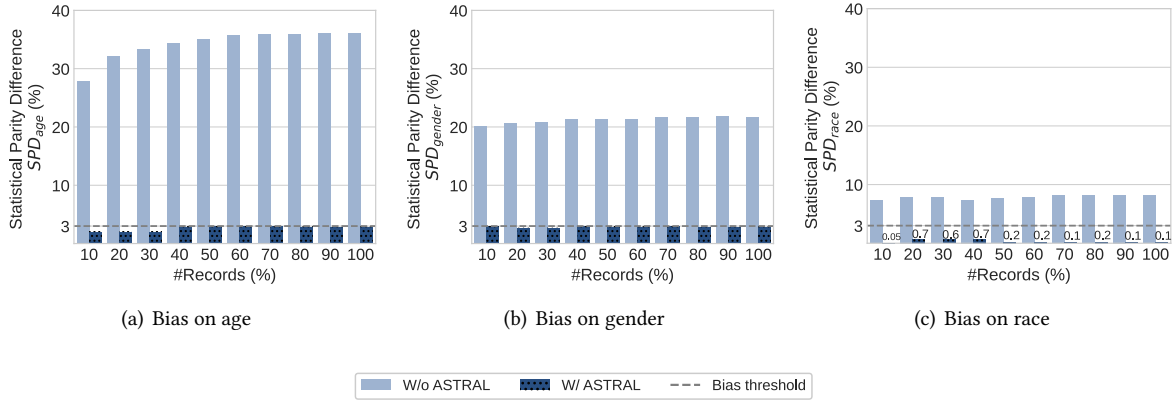


Fig. 9. ASTRAL's scalability with regard to the data size scale

20-client setup data, and so on, until reaching the 100-client setup, which contains the whole KDD train set. For each client setup, we used the same data distribution, experimental setup, and hyper-parameters for ASTRAL and we set the bias threshold to be 3%. We report how ASTRAL performs with the number of clients compared to FedAvg baseline in Fig. 10, showing SPD on age, gender and race sensitive attributes. We observe that for every number of clients, ASTRAL succeeds at making the global model respect the bias threshold.

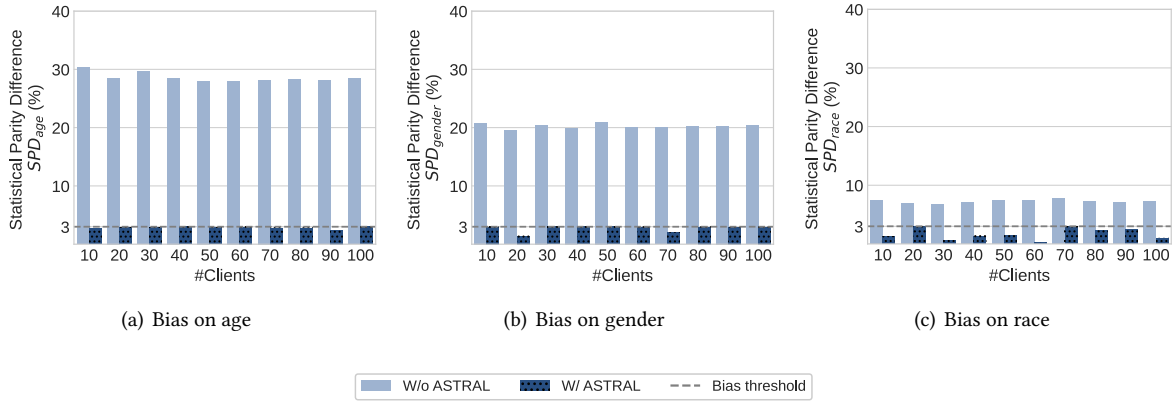


Fig. 10. ASTRAL's scalability with regard to the number of clients scale

5.10 FL System Scalability with Regard to Client Selection Ratios

In addition to evaluating ASTRAL's performance when varying the number of FL clients, we assess ASTRAL's robustness under different FL client participation ratios. We gradually increase the ratio of selected clients participating in a FL round, from 30% to 100% out of 10 FL clients, using ARS dataset. The corresponding results for model accuracy and model bias measured through SPD are presented in Fig. 11. The results show that ASTRAL

consistently ensures that the global model adheres to the bias threshold while maximizing model accuracy with different client selection ratios. However, we observe that when 3 FL clients are selected, the model accuracy decreases compared to FedAvg. This can be explained by the fact that having only 3 FL participating clients reduces the aggregation weights search space for ASTRAL and, thus, impacts accuracy.

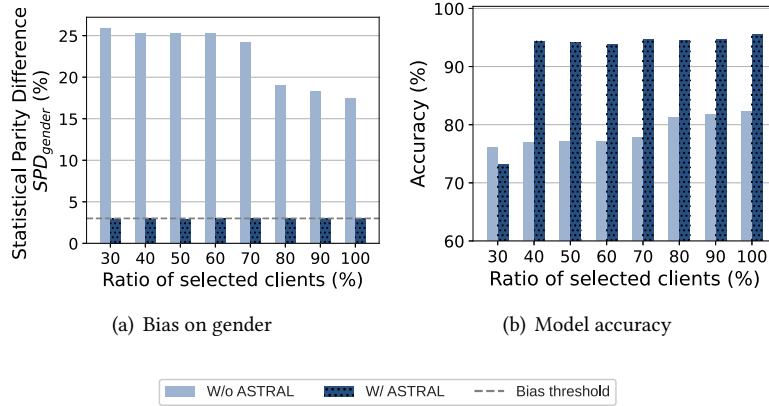


Fig. 11. ASTRAL's scalability with regard to client selection ratios

5.11 Robustness of Bias Mitigation with Respect to Clients' Data Heterogeneity

To evaluate a FL solution, a comprehensive benchmark must investigate its behavior in practical FL settings, in the presence of data heterogeneity. In this section, we evaluate how ASTRAL performs under different clients' data heterogeneity degrees. To achieve this, we use KDD dataset. We generate several FL setups using Dirichlet process. We vary the α parameter of the Dirichlet process to control how identical the clients are in each setup. Smaller α values result in a more heterogeneous data distribution among clients, while larger values lead to more similar and uniform clients as shown in Fig. 12. We evaluate ASTRAL under different levels of heterogeneity, with $\alpha \in [0.1, 0.2, 0.5, 1]$ and $\alpha \rightarrow \infty$. We consider for bias mitigation the three sensitive attributes age, gender, and race, and the SPD metric.

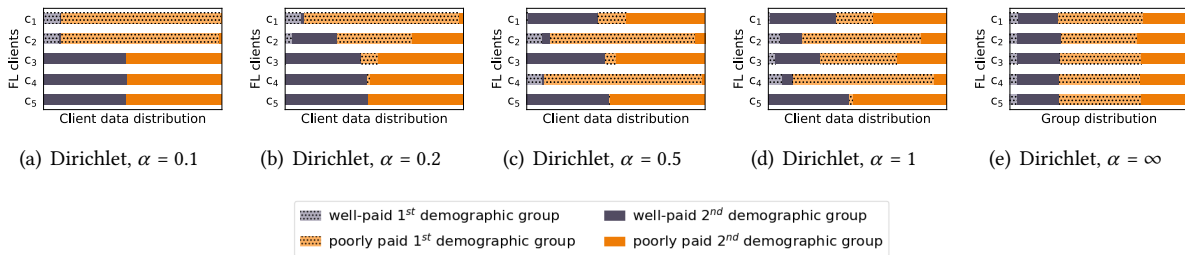


Fig. 12. Different non-IID client data distributions – More heterogeneous distribution from left to right – KDD dataset with 5 FL clients

Fig. 13 shows the SPD values of both FedAvg and ASTRAL for the considered Dirichlet parameters, for age, gender, and race. ASTRAL consistently satisfies the 3% bias constraint at different levels of heterogeneity. These findings demonstrate that ASTRAL remains unaffected by varying data heterogeneity levels and effectively addresses bias concerns.

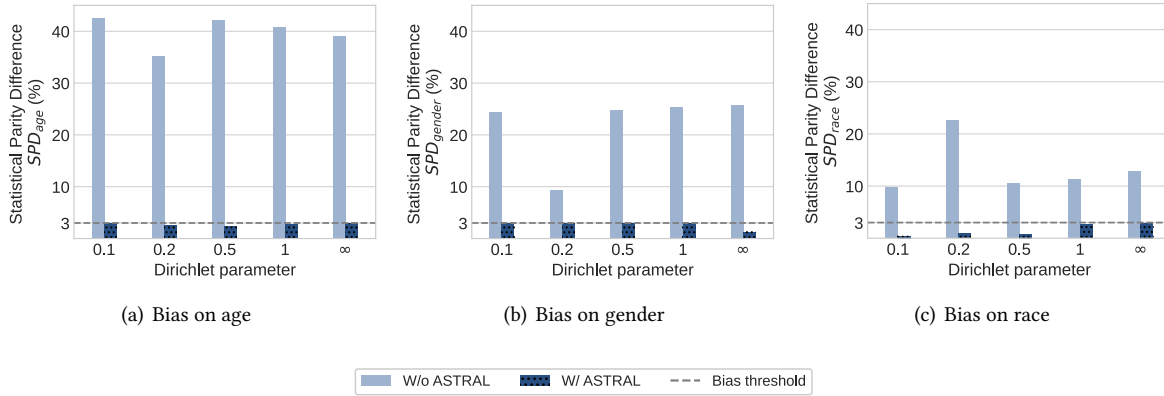


Fig. 13. Effectiveness of bias mitigation under different levels of clients' data heterogeneity

5.12 Impact of Data Distribution of Proxy Dataset on Bias Mitigation

In order to investigate the impact of the proxy dataset's data distribution on the proposed method, we evaluate the performance of ASTRAL with three distinct proxy datasets. These include one dataset that is representative of the training distribution, and two others deviating from the training distribution. We quantify the dissimilarity between the proxy datasets and the training data using the 1-Wasserstein distance [68], a proven metric for comparing data distributions [21, 27, 67]. It explores dissimilarity by optimally mapping one distribution to another. The higher the distance value, the further apart the two distributions are. The range of the distance is influenced by the inherent characteristics of the data.

We show the evaluation results for ARS dataset in Fig. 14, where we present the Wasserstein distances between the considered proxy datasets distributions and the training data distribution, the model bias (SPD) for each considered case, and the model accuracy. We observe that ASTRAL is able to successfully mitigate bias while achieving high accuracy for the different considered cases. Furthermore, we observe that when the proxy dataset's distribution is representative of the training data, ASTRAL achieves the highest accuracy as it can leverage fully the knowledge gained from the local models trained on the training distribution to construct the global model. As the proxy dataset diverges more from the training data distribution, the accuracy of the model obtained by ASTRAL declines. This is because it becomes more challenging to leverage the knowledge gained from one distribution (the training distribution) to produce a high-quality overall model with regard to a different distribution. This problem, that we consider orthogonal to our work, is known in ML literature as *data shift* [50]. There are several strategies and techniques that can be employed to mitigate its impact such as data augmentation, data monitoring, regularization techniques, *etc.* [50].

5.13 Stability of Bias Mitigation

In this section, we analyze the stability of ASTRAL at ensuring the bias threshold constraint through FL rounds. For that, we perform bias mitigation using ASTRAL applied on the SPD metric on all datasets. To be able to

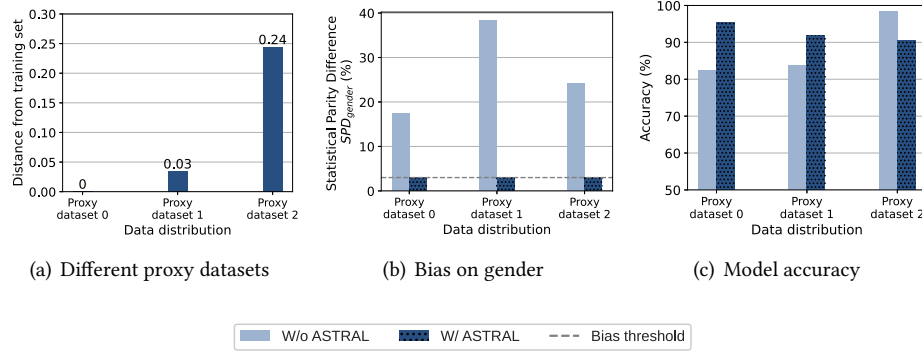


Fig. 14. Impact of different data distributions of proxy datasets on bias mitigation

compare ASTRAL's stability to all the state of art methods at once, including FairFed* and FCFL, we report results from experiments considering one sensitive attribute for each dataset: *gender* for DC, ARS, CelebA and MobiAct, *age* for KDD and Adult and *race* for MEPS. The results for ARS, MobiAct, CelebA, KDD, DC, Adult and MEPS are reported respectively in Fig. 15(a), Fig. 15(b), Fig. 15(c), Fig. 15(d), Fig. 15(e), Fig. 15(g), and Fig. 15(f). We can observe that for all the datasets, ASTRAL maintains the bias under the threshold constraint for all the FL rounds, meaning that ASTRAL's pipeline is able to find feasible solutions that ensures the bias constraint starting from the first round. FairFL* fails as its restrictive policy makes it ignore all clients, except in MobiAct, CelebA and Adult where it selects some clients, but resulting in a global model with less accuracy. We explained this observation in §5.4. The FairFed* policy reduces bias for the different datasets but only ensures the threshold for ARS, MobiAct and temporary for DC and CelebA. Moreover finding the proper aggregation weights that reduce bias in this method is a slow process compared to ASTRAL. As for FCFL, it is not compatible with the use of ResNet applied to CelebA dataset and MLP used with MobiAct. Consequently, the results are unavailable for this method on these particular datasets. It succeeds in guaranteeing the bias constraint only for MEPS and Adult. We observe on MEPS that in the first dozens rounds, FCFL struggles to keep the bias constraint in check and only stabilizes after 38 rounds. We observe a similar behavior on the Adult dataset for FCFL, only reaching stability in bias constraint respect after 300 rounds. In general, FCFL needs more FL rounds to stabilize. This is because FCFL relies on a gradient-based algorithm that modifies the global model after each local iteration at the clients' level, trying to ensure the optimization objective in the long run. For KDD and DC, FCFL is not able to reach the predefined threshold. For ARS, FCFL is not able to stabilize while maintaining the bias objective. As the authors mentioned in their work [11], FCFL's objective is to ensure the bias threshold for each client at its local level. This implies that, when data among clients is heterogeneous, and the global distribution is not represented by the individual clients distributions, than even if FCFL mitigates bias locally it does not mitigate it with regard to the global distribution. Meanwhile, ASTRAL manages to directly reach and stabilizes bias constraint within the first FL round on all datasets.

5.14 Cost Analysis

In order to assess how practical a given bias mitigation mechanism is, we evaluate its cost in terms of the number of FL rounds necessary to meet a bias constraint and maximize accuracy, as well as in terms of the amount of data exchanged over the network by the FL bias mitigation protocol and its total execution time. We evaluate the different bias mitigation mechanisms (FairFL*, FairFed*, FCFL, ASTRAL) constraining SPD below 3%, on all

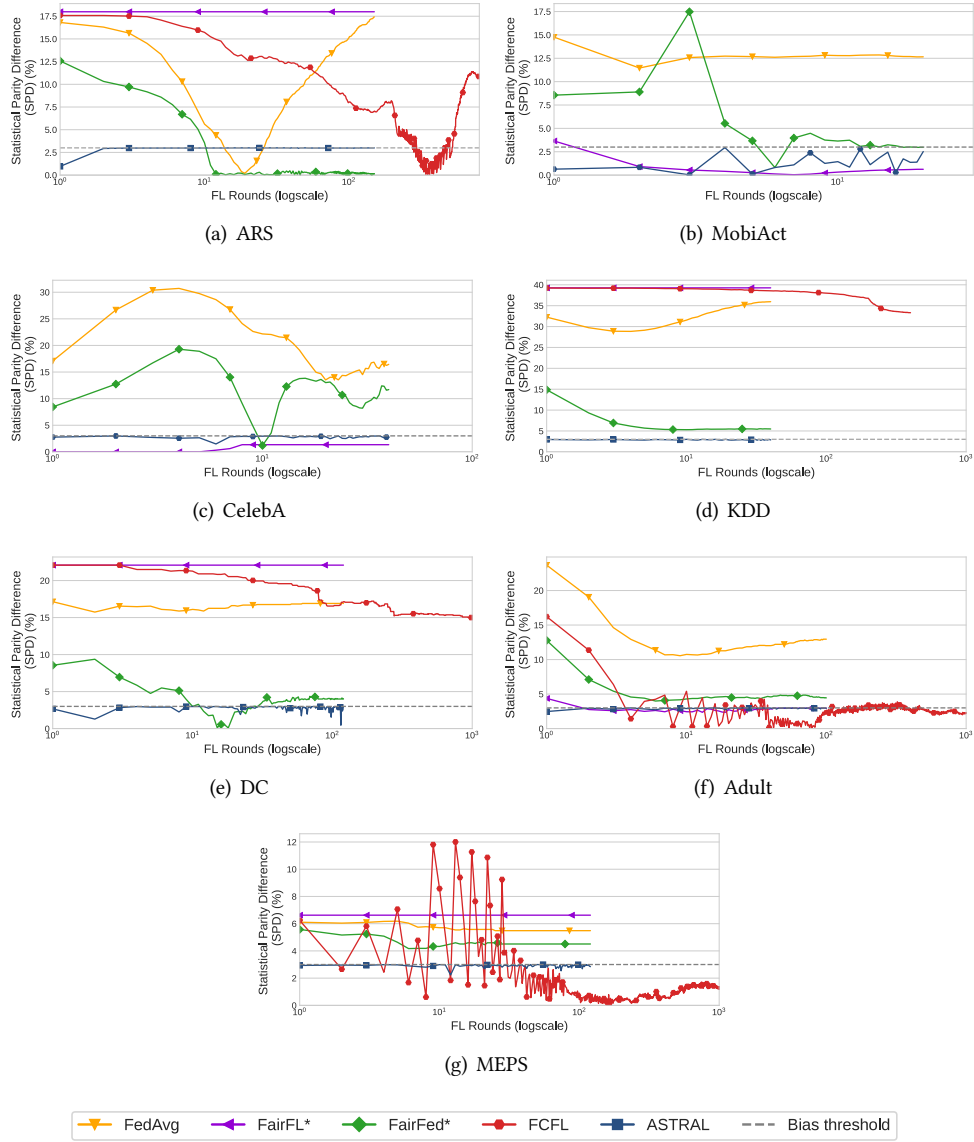


Fig. 15. Stability of bias mitigation

the datasets KDD, DC, Adult, MEPS, ARS, CelebA, and MobiAct. In order to be able to consider FCFL in this comparative evaluation, and since the former only handles a single sensitive attribute, we evaluate the cost of all systems with bias mitigation for a single sensitive attribute. As mentioned previously, we have not been able to conduct experiments with FCFL for MobiAct and CelebA due to its lack of support to ResNet and MLP models.

Table 6. Cost of bias mitigation. We show results for the max (100^{th}), the 95^{th} and the 90^{th} percentile of accuracy

Dataset	System	100^{th}			95^{th}			90^{th}		
		FL Round	Exchanged data (KB)	Accuracy	FL Round	Exchanged data (KB)	Accuracy	FL Round	Exchanged data (KB)	Accuracy
ARS	FairFL*	X	X	X	X	X	X	X	X	X
ARS	FairFed*	48	38	93	16	13	90	11	9	88
ARS	FCFL	519	649	83	X	X	X	X	X	X
ARS	ASTRAL	26	20	95	1	0.8	92	1	0.8	92
MEPS	FairFL*	X	X	X	X	X	X	X	X	X
MEPS	FairFed*	X	X	X	X	X	X	X	X	X
MEPS	FCFL	12,636	80,357	86	79	502	81	50	318	78
MEPS	ASTRAL	1	4	86	1	4	86	1	4	86
Adult	FairFL*	31	245	79	5	39	78	2	16	76
Adult	FairFed*	X	X	X	X	X	X	X	X	X
Adult	FCFL	7,691	91,631	82	111	3,534	78	74	882	74
Adult	ASTRAL	13	103	80	1	8	79	1	8	79
DC	FairFL*	X	X	X	X	X	X	X	X	X
DC	FairFed*	27	211	71	X	X	X	25	195	70
DC	FCFL	X	X	X	X	X	X	X	X	X
DC	ASTRAL	116	906	78	2	16	76	1	8	73
KDD	FairFL*	X	X	X	X	X	X	X	X	X
KDD	FairFed*	X	X	X	X	X	X	X	X	X
KDD	FCFL	X	X	X	X	X	X	X	X	X
KDD	ASTRAL	6	93	75	1	15	74	1	15	74
MobiAct	FairFL*	2	590	76	X	X	X	X	X	X
MobiAct	FairFed*	X	X	X	X	X	X	X	X	X
MobiAct	FCFL	X	X	X	X	X	X	X	X	X
MobiAct	ASTRAL	5	1,475	94	1	295	89	1	295	89
CelebA	FairFL*	8	3 GB	54	X	X	X	X	X	X
CelebA	FairFed*	10	3 GB	76	X	X	X	X	X	X
CelebA	FCFL	X	X	X	X	X	X	X	X	X
CelebA	ASTRAL	33	11 GB	89	10	3 GB	85	7	2 GB	80

We show the results in Table 6. The column 100^{th} of Table 6 presents the measurements collected when the bias mitigation methods reach their highest accuracy while maintaining bias under the predefined threshold. The column 95^{th} (respectively 90^{th}) presents the measurements collected when a method achieves a required accuracy defined as 95% (respectively 90%) of the maximum accuracy achieved for the dataset. For example, the maximum accuracy achieved for the dataset DC is 78%, by ASTRAL. the column 95^{th} presents the measurements of all methods applied on DC when they achieve 95% of the maximum accuracy, *i.e.*, 74%. The presence of a cross mark indicates that a method is not able to mitigate bias while achieving the required accuracy.

Overall, FairFL* is able to meet the bias constraint only with Adult, FairFed* meets it for ARS and for DC, FCFL meets it with Adult and MEPS, and ARS, and ASTRAL achieves the bias objective with all the datasets. For KDD, ASTRAL reaches the bias objective while scoring its highest accuracy at FL round 6, and scores more than 95% of its maximum accuracy starting at FL round 1. The amount of exchanged data is between 15 KB and 93 KB. In contrast, FairFL* and FCFL are not able to reach the bias objective for that dataset. For the DC dataset, ASTRAL takes 116 rounds to reach its highest accuracy (78%), and only 2 rounds to reach the accuracy (76%). In general, it ensures higher accuracy compared to FairFed*. For Adult, ASTRAL reaches its maximum accuracy while fulfilling the bias objective at FL round 13, with 103 KB of exchanged data. This is far better than FairFL* and FCFL which score their maximum accuracy only after respectively 31 and 7, 691 FL rounds, and after respectively 245 KB and 91, 631 KB of exchanged data. If FCFL is able to achieve the bias objective at earlier rounds, it is at the expense of a lower accuracy (74%). The trend is even stronger with MEPS for which ASTRAL scores its maximum accuracy at FL round 1, while FCFL scores the same maximum accuracy after its 12, 636th FL round. ASTRAL needs 4 KB of exchanged data, while FCFL uses 80, 357 KB. As explained in §5.13, FCFL takes typically more FL rounds to stabilize because it requires from the clients to communication with the server after each local iteration at the client's level, via a gradient-based algorithm. In other terms, one FL round in FCFL comprises necessary one local

epoch of training. For the ARS dataset, we observe that FairFL* does not achieve the bias constraint due to the selection process which fails from the first round. FairFed* reaches the bias constraint starting from round 11, and achieves the best accuracy while respecting the bias constraint at round 48. FCFL reaches the bias constraint only after more than hundred rounds, and achieves the best accuracy while respecting the bias constraint only at round 519. ASTRAL is able to ensure the bias objective directly from the first round with an accuracy of 92%. Moreover, it achieves the best accuracy of 95% compared to the other methods while respecting the bias threshold at round 26. For both MobiAct and CelebA, ASTRAL maintains the bias objective while achieving the highest accuracies compared to FairFL* and FairFed*. Moreover, ASTRAL outperforms FairFL* and FairFed* in terms of accuracy starting from early rounds, 1st round for MobiAct and 7th for CelebA. In summary, these experiments show that ASTRAL is a practical FL bias mitigation solution, and is more efficient than state-of-the-art solutions in terms of maximizing accuracy, communication rounds, and amount of exchanged data.

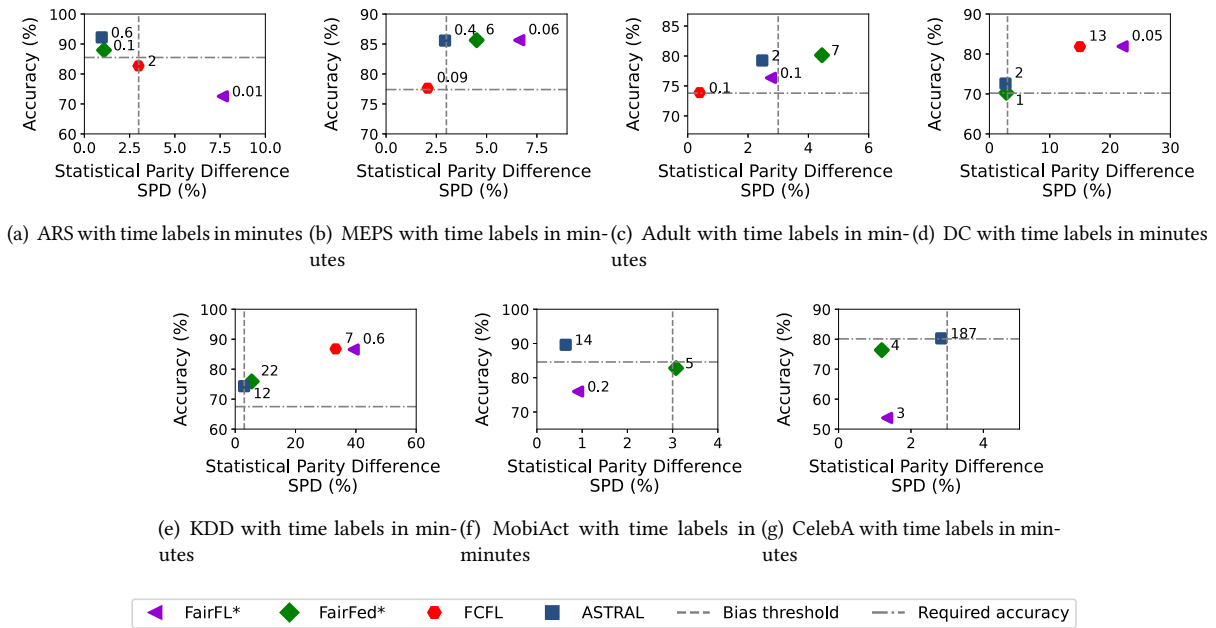


Fig. 16. Trade-off between execution time, accuracy and bias

In the following we are interested in analyzing the trade-offs between accuracy, bias, and execution time for the considered bias mitigation methods. The trade-offs measurements are presented for all the datasets in Fig. 16. Each sub-figure presents the accuracy and bias of the different bias mitigation methods for a given dataset, at the round where the methods attain 90% of the highest observed accuracy (see column 90th of Table 6). Each sub-figure also presents, through labels, the total execution time to achieve the required objective of a method including the clients' training time, the models' transmission time, and the necessary time for models' aggregation at the server level.

We can observe from Fig. 16 that ASTRAL is the only method that consistently occupies the upper-left square of the graphs, limited by the required accuracy threshold at the bottom and the predefined bias threshold on the right. This shows that ASTRAL consistently attains the highest accuracies while ensuring bias mitigation.

In terms of execution time, ASTRAL's performance ranges from just a few seconds to around 12 minutes, except for the case of CelebA, where ASTRAL's execution time stands out. In this particular scenario, ASTRAL's overall execution time is relatively higher compared to that of the other datasets. This difference can be attributed to the overhead involved in training a complex model like ResNet on the large image dataset of CelebA. Moreover, specifically for the CelebA case, we noted that ASTRAL's algorithm requires additional time to identify the optimal aggregation weight. This phenomenon is attributed to the characteristics of the model and data employed in this particular use case. Nevertheless, as mentioned in our earlier discussions, ASTRAL's execution time can be optimized through the fine-tuning of its hyperparameters. Furthermore, making use of parallelization, such as employing multi-core machines at the server level, greatly benefits the execution time of the Differential Evolution algorithm [13].

6 CONCLUSION

We presented ASTRAL, a novel FL aggregation method designed to mitigate bias for FL in ubiquitous systems. ASTRAL finds the best combination of FL clients' local model parameters in order to obtain a global FL model ensuring that bias remains below a given threshold, while keeping model accuracy as high as possible. ASTRAL allows for a per-use-case bias threshold configuration, a practical approach that can be easily deployed in diverse ubiquitous computing applications. ASTRAL handles the presence of multiple sensitive attributes in data and mitigates bias for all of them simultaneously. It also supports by design bias mitigation for any bias metric, adapting bias mitigation to diverse practical use cases. Through our extensive empirical study of ASTRAL with seven real-world datasets, we assessed its performance in terms of bias and accuracy objectives, scalability, robustness against data heterogeneity, and low communication cost. The results show that ASTRAL's aggregation approach that provides the best possible aggregation weights with regard to our objectives leads to effective bias mitigation and model accuracy. We intend to further extend this work as follows. We plan to extend the multi-objective approach of ASTRAL to consider additional elements on top of bias and accuracy, including privacy, robustness, and efficiency [34]. As the use of FL gains more traction across several applications, it becomes apparent that mitigating bias and optimizing accuracy may not be sufficient. Privacy must also be taken into account to protect sensitive information. Similarly, robustness is essential for ensuring that FL models are not susceptible to attacks [48]. Incorporating these additional aspects into the multi-objective approach results in effective and reliable FL models that can be trusted to be leveraged for ubiquitous computing systems.

ACKNOWLEDGMENTS

This work was partially supported by French ANR project ByBLoS (ANR-20-CE25-0002-01). Experiments presented in this paper were carried out using the Grid'5000 testbed, supported by a scientific interest group hosted by Inria and including CNRS, RENATER, and several universities as well as other organizations². We would also like to extend our thanks to Paul Wawszczyk, who participated in the project as an intern.

REFERENCES

- [1] Annie Abay, Yi Zhou, Nathalie Baracaldo, Shashank Rajamoni, Ebube Chuba, and Heiko Ludwig. 2020. Mitigating Bias in Federated Learning. arXiv:2012.02447
- [2] Simeon Okechukwu Ajakwe, Rubina Arkter, Love Allen Chijioke Ahakonye, Dong-Seong Kim, and Jae-Min Lee. 2021. Real-Time Monitoring of COVID-19 Vaccination Compliance: A Ubiquitous IT Convergence Approach. In *2021 International Conference on Information and Communication Technology Convergence (ICTC)*. IEEE, Jeju Island, Korea, 440–445.
- [3] P Apoorva., H.C Impana., S.L Siri., M.R Varshitha., and B Ramesh. 2019. Automated Criminal Identification by Face Recognition Using Open Computer Vision Classifiers. In *2019 3rd International Conference on Computing Methodologies and Communication (ICCMC)*. IEEE, Erode, Tamil Nadu, India, 775–778.

²See <https://www.grid5000.fr>

- [4] Stephan Böhm and Susanne J. Niklas. 2012. Mobile Recruiting: Insights from a Survey among German HR Managers. In *Proceedings of the 50th Annual Conference on Computers and People Research* (Milwaukee, Wisconsin, USA) (SIGMIS-CPR '12). Association for Computing Machinery, New York, NY, USA, 117–122.
- [5] Emanuela Bran, Elena Bautu, Dragos Florin Sburlan, Crenguta Madalina Puchianu, and Dorin Mircea Popovici. 2021. Ubiquitous Computing: Driving in the Intelligent Environment. *Mathematics* 9, 21 (2021), 2649.
- [6] Hong-You Chen and Wei-Lun Chao. 2021. FedBE: Making Bayesian Model Ensemble Applicable to Federated Learning. In *9th International Conference on Learning Representations, ICLR 2021, May 3-7, 2021*. OpenReview.net, Virtual Event, Austria, .
- [7] Yiqiang Chen, Xin Qin, Jindong Wang, Chaohui Yu, and Wen Gao. 2020. FedHealth: A Federated Transfer Learning Framework for Wearable Healthcare. *IEEE Intelligent Systems* 35, 4 (2020), 83–93.
- [8] Zhengyu Chen, Teng Xiao, and Kun Kuang. 2022. BA-GNN: On Learning Bias-Aware Graph Neural Network. In *38th IEEE International Conference on Data Engineering, (ICDE 2022)*. IEEE, Kuala Lumpur, Malaysia, 3012–3024.
- [9] Lingyang Chu, Lanjun Wang, Yanjie Dong, Jian Pei, Zirui Zhou, and Yong Zhang. 2021. FedFair: Training Fair Models In Cross-Silo Federated Learning. arXiv:2109.05662
- [10] Steven Cohen. 2003. Design Strategies and Innovations in the Medical Expenditure Panel Survey. *Medical care* 41 (08 2003), III5–III12.
- [11] Sen Cui, Weishen Pan, Jian Liang, Changshui Zhang, and Fei Wang. 2021. Addressing Algorithmic Disparity and Performance Inconsistency in Federated Learning. *Advances in Neural Information Processing Systems* 34 (2021), 26091–26102.
- [12] Sen Cui, Weishen Pan, Jian Liang, Changshui Zhang, and Fei Wang. 2021. *Repository of FCFL Software for Bias Mitigation in Federated Learning*. Institute for Artificial Intelligence, Tsinghua University THUAI). Retrieved November 14, 2022 from <https://github.com/cuis15/FCFL>
- [13] Swagatam Das and Ponnuthurai Nagarathnam Suganthan. 2011. Differential Evolution: A Survey of the State-of-the-Art. *IEEE Transactions on Evolutionary Computation* 15, 1 (2011), 4–31.
- [14] Naishadh K Dave and Vanaraj B Vaghela. 2009. Vehicular Traffic Control: A Ubiquitous Computing Approach. In *Contemporary Computing: Second International Conference, IC3 2009, August 17-19, 2009. Proceedings 2*. Springer, Noida, India, 336–348.
- [15] Michele Donini, Luca Oneto, Shai Ben-David, John Shawe-Taylor, and Massimiliano Pontil. 2018. Empirical Risk Minimization Under Fairness Constraints. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems (NeurIPS'18)*. Neural Information Processing Systems, Montréal, Canada, 2796–2806.
- [16] Wei Du, Depeng Xu, Xintao Wu, and Hanghang Tong. 2021. *Fairness-Aware Agnostic Federated Learning*. SIAM, Virtual Event, 181–189.
- [17] Dheeru Dua and Casey Graff. 2017. UCI Machine Learning Repository. <http://archive.ics.uci.edu/ml>
- [18] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. 2012. Fairness Through Awareness. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference* (Cambridge, Massachusetts) (ITCS '12). Association for Computing Machinery, New York, NY, USA, 214–226.
- [19] Departments and Agencies of the Federal Government 2017. *Equal Credit Opportunity Act*. Departments and Agencies of the Federal Government. Retrieved January 24, 2022 from <https://www.ecfr.gov/current/title-12/chapter-II/subchapter-A/part-202/section-202.2>
- [20] Yahya H. Ezzeldin, Shen Yan, Chaoyang He, Emilio Ferrara, and Amir Salman Avestimehr. 2023. FairFed: Enabling Group Fairness in Federated Learning. In *Thirty-Seventh AAAI Conference on Artificial Intelligence, AAAI 2023, Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence, IAAI 2023, Thirteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2023, February 7-14, 2023*, Brian Williams, Yiling Chen, and Jennifer Neville (Eds.). AAAI Press, Washington, DC, USA, 7494–7502.
- [21] Wei Fan, Kunpeng Liu, Rui Xie, Hao Liu, Hui Xiong, and Yanjie Fu. 2021. Fair Graph Auto-Encoder for Unbiased Graph Representations with Wasserstein Distance. In *IEEE International Conference on Data Mining, ICDM 2021, December 7-10, 2021*, James Bailey, Pauli Miettinen, Yun Sing Koh, Dacheng Tao, and Xindong Wu (Eds.). IEEE, Auckland, New Zealand, 1054–1059.
- [22] Michael Feldman, Sorelle A. Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and Removing Disparate Impact. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (Sydney, NSW, Australia) (KDD '15). Association for Computing Machinery, New York, NY, USA, 259–268.
- [23] Michael R. Garey and David S. Johnson. 1979. Computers and Intractability: A Guide to the Theory of NP-Hardness.
- [24] Manolis Georgioudakis and Vagelis Plevris. 2020. A Comparative Study of Differential Evolution Variants in Constrained Structural Optimization. *Frontiers in Built Environment* 6 (2020), 102.
- [25] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of Opportunity in Supervised Learning. *Advances in Neural Information Processing Systems* 29 (2016), 3315–3323.
- [26] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016, June 27-30, 2016*. IEEE Computer Society, Las Vegas, NV, USA, 770–778.
- [27] Xing He, Changgen Peng, and Weijie Tan. 2023. Fast and Accurate Deep Leakage from Gradients Based on Wasserstein Distance. *International Journal of Intelligent Systems* 2023 (2023), 1–12.
- [28] Kevin Hsieh, Amar Phanishayee, Onur Mutlu, and Phillip B. Gibbons. 2020. The Non-IID Data Quagmire of Decentralized Machine Learning. In *Proceedings of the 37th International Conference on Machine Learning, ICML 2020 (Proceedings of Machine Learning Research, Vol. 119)*. PMLR, Virtual Event, 4387–4398.

- [29] Tzu-Ming Harry Hsu, Hang Qi, and Matthew Brown. 2019. Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification. *arXiv:1909.06335*
- [30] Zeou Hu, Kiarash Shaloudegi, Guojun Zhang, and Yaoliang Yu. 2022. Federated Learning Meets Multi-Objective Optimization. *IEEE Transactions on Network Science and Engineering* 9, 4 (2022), 2039–2051.
- [31] Legal Information Institute. 1978. 29 CFR 1607.4 - Information on Impact. <https://www.law.cornell.edu/cfr/text/29/1607.4>
- [32] Ji Chu Jiang, Burak Kantarci, Sema Oktug, and Tolga Soyata. 2020. Federated Learning in Smart City Sensing: Challenges and Opportunities. *Sensors* 20, 21 (2020), 6230.
- [33] Yaochu Jin. 2005. A Comprehensive Survey of Fitness Approximation in Evolutionary Computation. *Soft computing* 9, 1 (2005), 3–12.
- [34] Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista A. Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, Rafael G. L. D'Oliveira, Hubert Eichner, Salim El Rouayheb, David Evans, Josh Gardner, Zachary Garrett, Adrià Gascón, Badi Ghazi, Phillip B. Gibbons, Marco Gruteser, Zaid Harchaoui, Chaoyang He, Lie He, Zhouyuan Huo, Ben Hutchinson, Justin Hsu, Martin Jaggi, Tara Javidi, Gauri Joshi, Mikhail Khodak, Jakub Konečný, Aleksandra Korolova, Farinaz Koushanfar, Sanmi Koyejo, Tancrede Lepoint, Yang Liu, Prateek Mittal, Mehryar Mohri, Richard Nock, Ayfer Özgür, Rasmus Pagh, Hang Qi, Daniel Ramage, Ramesh Raskar, Mariana Raykova, Dawn Song, Weikang Song, Sebastian U. Stich, Ziteng Sun, Ananda Theertha Suresh, Florian Tramèr, Praneeth Vepakomma, Jianyu Wang, Li Xiong, Zheng Xu, Qiang Yang, Felix X. Yu, Han Yu, and Sen Zhao. 2021. Advances and Open Problems in Federated Learning. *Found. Trends Mach. Learn.* 14, 1-2 (2021), 1–210.
- [35] Toshihiro Kamishima, Shotaro Akaho, Hideki Asoh, and Jun Sakuma. 2012. Fairness-Aware Classifier With Prejudice Remover Regularizer. In *Machine Learning and Knowledge Discovery in Databases*, Peter A. Flach, Tijl De Bie, and Nello Cristianini (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 35–50.
- [36] Ron Kohavi. 1996. Scaling Up the Accuracy of Naive-Bayes Classifiers: A Decision-Tree Hybrid. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (Portland, Oregon) (KDD'96)*. AAAI Press, Portland, Oregon, USA, 202–207.
- [37] Tian Li, Maziar Sanjabi, Ahmad Beirami, and Virginia Smith. 2019. Fair Resource Allocation in Federated Learning. *arXiv:1905.10497*
- [38] Youpeng Li, Xuyu Wang, and Lingling An. 2023. Hierarchical Clustering-Based Personalized Federated Learning for Robust and Fair Human Activity Recognition. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 1, Article 20 (mar 2023), 38 pages.
- [39] Zexi Li, Tao Lin, Xinyi Shang, and Chao Wu. 2023. Revisiting Weighted Aggregation in Federated Learning with Neural Networks. *arXiv:2302.10911*
- [40] Tao Lin, Lingjing Kong, Sebastian U. Stich, and Martin Jaggi. 2020. Ensemble Distillation for Robust Model Fusion in Federated Learning. In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020 (NeurIPS'20, Vol. 33)*, Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin (Eds.). Neural Information Processing Systems, Virtual, 2351–2363.
- [41] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. 2015. Deep Learning Face Attributes in the Wild. In *2015 IEEE International Conference on Computer Vision, ICCV 2015, December 7-13, 2015*. IEEE Computer Society, Santiago, Chile, 3730–3738.
- [42] Lingjuan Lyu, Xinyi Xu, Qian Wang, and Han Yu. 2020. Collaborative Fairness in Federated Learning. *Federated Learning: Privacy and Incentive* 12500 (2020), 189–204.
- [43] Trisha Mahoney, Kush Varshney, and Michael Hind. 2020. *AI Fairness*. O'Reilly Media, Incorporated, .
- [44] John McCall. 2005. Genetic Algorithms for Modelling and Optimisation. *Journal of computational and Applied Mathematics* 184, 1 (2005), 205–222.
- [45] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera Y. Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research, Vol. 54)*, Aarti Singh and Jerry Zhu (Eds.). PMLR, Fort Lauderdale Florida, USA, 1273–1282.
- [46] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2022. A Survey on Bias and Fairness in Machine Learning. *Comput. Surveys* 54, 6 (2022), 115:1–115:35.
- [47] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2022. A Survey on Bias and Fairness in Machine Learning. *Comput. Surveys* 54, 6 (2022), 115:1–115:35.
- [48] Aghiles Ait Messaoud, Sonia Ben Mokhtar, Vlad Nitu, and Valerio Schiavoni. 2022. Shielding Federated Learning Systems Against Inference Attacks with ARM TrustZone. In *Proceedings of the 23rd ACM/IFIP International Middleware Conference*. ACM, Quebec, QC, Canada, 335–348.
- [49] Mehryar Mohri, Gary Sivek, and Ananda Theertha Suresh. 2019. Agnostic Federated Learning. In *International Conference on Machine Learning*. PMLR, California, USA , 4615–4625.
- [50] Jose G Moreno-Torres, Troy Raeder, Rocio Alaiz-Rodríguez, Nitesh V Chawla, and Francisco Herrera. 2012. A Unifying View on Dataset Shift in Classification. *Pattern recognition* 45, 1 (2012), 521–530.
- [51] Fionn Murtagh. 1991. Multilayer Perceptrons for Classification and Regression. *Neurocomputing* 2, 5-6 (1991), 183–197.
- [52] Dinh C. Nguyen, Quoc-Viet Pham, Pubudu N. Pathirana, Ming Ding, Aruna Seneviratne, Zihuai Lin, Octavia A. Dobre, and Won-Joo Hwang. 2021. Federated Learning for Smart Healthcare: A Survey. *arXiv:2111.08834 [cs.LG]*

- [53] Eirini Ntoutsis, Pavlos Fafalios, Ujwal Gadiraju, Vasileios Iosifidis, Wolfgang Nejdl, Maria-Esther Vidal, Salvatore Ruggieri, Franco Turini, Symeon Papadopoulos, Emmanouil Krasanakis, et al. 2020. Bias in Data-Driven Artificial Intelligence Systems – An Introductory Survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 10, 3 (2020), e1356.
- [54] Sharnil Pandya, Gautam Srivastava, Rutvij Jhaveri, M Rajasekhara Babu, Sweta Bhattacharya, Praveen Kumar Reddy Maddikunta, Spyridon Mastorakis, Md Jalil Piran, and Thippa Reddy Gadekallu. 2023. Federated Learning for Smart Cities: A Comprehensive Survey. *Sustainable Energy Technologies and Assessments* 55 (2023), 102987.
- [55] Drazen Prelec. 1998. The Probability Weighting Function. *Econometrica* 66, 3 (1998), 497–527.
- [56] Luis Rios and Nikolaos Sahinidis. 2013. Derivative-Free Optimization: A Review of Algorithms and Comparison of Software Implementations. *Journal of Global Optimization* 56, 3 (July 2013), 1247–1293.
- [57] Luis Eduardo Romero, Parag Chatterjee, and Ricardo Luis Armentano. 2016. An IoT Approach for Integration of Computational Intelligence and Wearable Sensors for Parkinson’s Disease Diagnosis and Monitoring. *Health and Technology* 6 (2016), 167–172.
- [58] SCIPY 2023. *Differential Evolution - SciPy implementation*. SCIPY. Retrieved January 31, 2023 from https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.differential_evolution.html
- [59] Roberto L. Shinmoto Torres, Damith C. Ranasinghe, Qinfeng Shi, and Alanson P. Sample. 2013. Sensor Enabled Wearable RFID Technology for Mitigating the Risk of Falls Near Beds. In *2013 IEEE International Conference on RFID (RFID)*. IEEE, Johor Bahru, Malaysia, 191–198.
- [60] Michael W Sjoding, Robert P Dickson, Theodore J Iwashyna, Steven E Gay, and Thomas S Valley. 2020. Racial Bias in Pulse Oximetry Measurement. *New England Journal of Medicine* 383, 25 (2020), 2477–2478.
- [61] Rainer Storn and Kevin P. Price. 1997. Differential Evolution: A Simple and Efficient Adaptive Scheme for Global Optimization Over Continuous Spaces. *Journal of Global Optimization* 11 (1997), 341–359.
- [62] Dimitris Stripelis and José Luis Ambite. 2020. Accelerating Federated Learning in Heterogeneous Data and Computational Environments. arXiv:2008.11281
- [63] Weigao Su, Daibo Liu, Taiyuan Zhang, and Hongbo Jiang. 2022. Towards Device Independent Eavesdropping on Telephone Conversations with Built-in Accelerometer. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 5, 4, Article 177 (2022), 29 pages.
- [64] Emmanuel Munguia Tapia et al. 2004. Activity Recognition in the Home Using Simple and Ubiquitous Sensors. In *Pervasive Computing: Second International Conference, PERVASIVE 2004, April 21-23, 2004. Proceedings 2*. Springer, Linz/Vienna, Austria, 158–175.
- [65] EC 2021. *The Artificial Intelligence Act*. EC. Retrieved November 15, 2022 from <https://artificialintelligenceact.eu/>
- [66] U.S. House 2020. *The National AI Initiative Act*. U.S. House. Retrieved November 15, 2022 from <https://www.ai.gov/>
- [67] Alexander Tong, Guy Wolf, and Smita Krishnaswamy. 2022. Fixing Bias in Reconstruction-Based Anomaly Detection with Lipschitz Discriminators. *J. Signal Process. Syst.* 94, 2 (2022), 229–243.
- [68] SS Vallender. 1974. Calculation of the Wasserstein Distance Between Probability Distributions on the Line. *Theory of Probability & Its Applications* 18, 4 (1974), 784–786.
- [69] Paul Van der Laan. 2000. The 2001 Census in the Netherlands. In *Conference the Census of Population*. U.S. Department of Commerce, Washington, USA.
- [70] George Vavoulas, Charikleia Chatzaki, Thodoris Malliotakis, Matthew Padiaditis, and Manolis Tsiknakis. 2016. The Mobiact Dataset: Recognition of Activities of Daily Living Using Smartphones. In *International Conference on Information and Communication Technologies for Ageing Well and e-Health*, Vol. 2. SCITEPRESS, Rome, Italy, 143–151.
- [71] OSTP 2022. *Blueprint for an AI Bill of Rights*. OSTP. Retrieved January 24, 2023 from <https://www.whitehouse.gov/wp-content/uploads/2022/10/Blueprint-for-an-AI-Bill-of-Rights.pdf>
- [72] Yongkai Wu, Lu Zhang, and Xintao Wu. 2019. On Convexity and Bounds of Fairness-Aware Classification. In *The World Wide Web Conference* (San Francisco, CA, USA) (WWW ’19). Association for Computing Machinery, New York, NY, USA, 3356–3362.
- [73] Bangzhou Xin, Wei Yang, Yangyang Geng, Sheng Chen, Shaowei Wang, and Liusheng Huang. 2020. Private FL-GAN: Differential Privacy Synthetic Data Generation Based on Federated Learning. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Virtual Barcelona, 2927–2931.
- [74] Sofia Yfantidou, Marios Constantinides, Dimitris Spathis, Athena Vakali, Daniele Quercia, and Fahim Kawsar. 2023. Beyond Accuracy: A Critical Review of Fairness in Machine Learning for Mobile and Wearable Computing. arXiv:2303.15585
- [75] Han Yu, Zelei Liu, Yang Liu, Tianjian Chen, Mingshu Cong, Xi Weng, Dusit Niyato, and Qiang Yang. 2020. A Fairness-aware Incentive Scheme for Federated Learning. In *AIES ’20: AAAI/ACM Conference on AI, Ethics, and Society, February 7-8, 2020*, Annette N. Markham, Julia Powles, Toby Walsh, and Anne L. Washington (Eds.). ACM, New York, NY, USA, 393–399.
- [76] Xubo Yue, Maher Nouiehed, and Raed Al Kontar. 2023. Gifair-fl: A Framework for Group and Individual Fairness in Federated Learning. *INFORMS Journal on Data Science* 2, 1 (2023), 10–23.
- [77] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez Rogriguez, and Krishna P. Gummadi. 2017. Fairness Constraints: Mechanisms for Fair Classification. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research, Vol. 54)*, Aarti Singh and Jerry Zhu (Eds.). PMLR, Fort Lauderdale Florida, USA, 962–970.
- [78] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez Rogriguez, and Krishna P. Gummadi. 2019. Fairness Constraints: A Flexible Approach for Fair Classification. *Journal of Machine Learning Research* 20, 75 (2019), 75:1–75:42.

- [79] Yuchen Zeng, Hongxu Chen, and Kangwook Lee. 2021. Improving Fairness via Federated Learning. arXiv:2110.15545
- [80] Daniel Yue Zhang, Kou Ziyi, and Wang Dong. 2020. FairFL: A Fair Federated Learning Approach to Reducing Demographic Bias in Privacy-Sensitive Classification Models. In *Proceedings of the 2020 IEEE International Conference on Big Data, BigData2020*. IEEE, United States, 1051–1060.
- [81] Pengyuan Zhou, Hengwei Xu, Lik Hang Lee, Pei Fang, and Pan Hui. 2022. Are You Left Out? An Efficient and Fair Federated Learning for Personalized Profiles on Wearable Devices of Inferior Networking Conditions. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6, 2, Article 91 (jul 2022), 25 pages.